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MindfulFeed: A Stanford d.School Design Thinking Record for Ethical Social Media Design to Replace Dark Patterns and Protect Cognitive Well Being

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Abstract

This study synthesizes research on algorithmic personalization, dark patterns, and governance to produce and evaluate MindfulFeed, a pro-social social app prototype focused on preserving cognitive autonomy and psychological safety. Grounded design criteria translate four governance dimensions, accountability, transparency, oversight, and well-being outcomes, into concrete requirements that ban manipulative interface mechanisms and enforce symmetric optin/optout pathways. A Stanford d.School informed process combined empathic interviews, diary studies, expert cognitive walkthroughs, and iterative prototyping to generate three prototypes: a lowfidelity batch feed that replaces infinite scroll with session bounded consumption, a midfidelity algorithm control dashboard exposing tiered personalization and revocable consent, and a highfidelity interactive system implementing peritem explanations, fatigueaware nudges, and reversible profile adjustments. Testing protocols integrated automated dark pattern scans, task based usability, and qualitative coding mapped to governance pillars; iterative results reduced asymmetries in cancellation and consent flows, increased visibility of data use, and lowered reported cognitive strain. Ethical recommendation engine specifications emphasize data minimization, fairness audits, humanintheloop decision gates, and auditable logs. Monetization approaches were reconfigured to exclude urgency cues, confirm shaming, and hidden charges, aligning revenue paths with trust and regulatory resilience. Findings indicate that design constraints informed by taxonomies and governance metrics can materially reduce manipulative affordances while maintaining user value through transparent personalization. The work provides a replicable development record and a practical bridge between regulatory principles and product design for platforms aiming to prioritise long term user autonomy and psychological health.

1 Contextual Foundations of Ethical Social Media Design

1.1 Digital Engagement, Burnout, and WellBeing

Intensive participation in social media environments exposes users, particularly students and young adults, to multiple psychological risks that are closely related to burnout and diminished wellbeing. Cyberbullying represents a salient form of harmful digital engagement: anonymity in social platforms facilitates abusive behaviour without accountability, and victims frequently report depression, anxiety, social withdrawal, and in severe cases selfharm as consequences of sustained harassment. From an ethical perspective, such hostile interactions breach principles of respect, dignity, and nonmaleficence, while failures in platform moderation and redress mechanisms aggravate harm by leaving affected users without effective remedies. Platform design choices further shape the relationship between engagement and burnout. Many social media systems incorporate infinite scrolling, push notifications, and reward-based feedback elements such as likes, shares, and comments, intentionally engineered to prolong usage and capture attention through psychological reinforcement mechanisms (Dadage, 2026, p. 11). Evidence on AIenhanced environments shows that machine learning algorithms personalize notifications and content recommendations to maximize engagement, creating compulsive usage patterns and digital dependency that are especially prominent among adolescents and young adults (Ronald, 2025, p. 6). Excessive engagement of this type disrupts sleep, reduces academic productivity, and impairs offline

social interactions, aligning with burnoutrelated symptoms such as cognitive fatigue and social withdrawal (Dadage, 2026, p. 11; Ronald, 2025, p. 6). Research on digital wellbeing conceptualizes these outcomes within a multidimensional framework that links engagement intensity with psychological, cognitive, social, and physical states. Psychological consequences of AImediated environments include both positive and negative pathways: AIbased mood tracking and digital therapeutics can support individuals experiencing stress, anxiety, and depressive symptoms, yet algorithmically curated social media feeds that foreground idealized content and social comparison can heighten anxiety, loneliness, and emotional distress. Continuous exposure to such curated streams, particularly during formative years, undermines emotional stability and may accelerate burnout trajectories for heavy users (Ronald, 2025, p. 4). Social and cognitive dimensions compound these pressures. Recommendation algorithms help users form online communities that deliver emotional support and belonging, but they also restructure communication dynamics by prioritizing engagement metrics over meaningful interaction and may reduce facetoface communication (Ronald, 2025, p. 6). Studies of excessive digital engagement associate high usage with reduced attention control, emotional dysregulation, anxiety, depression, sleep disruption, diminished academic performance, workplace distraction, loneliness, and reduced subjective wellbeing, while still acknowledging countervailing benefits such as social connectedness and educational access (Padmaavathy et al., 2026, p. 6). This pattern supports a nonlinear view in which engagement is not inherently harmful, but becomes problematic when algorithmic reinforcement erodes autonomy and substitutes essential offline experiences (Padmaavathy et al., 2026, p. 6; Ronald, 2025, p. 6). Organizational contexts illustrate how continuous digital connectivity can culminate in burnout. AIsupported collaboration tools, virtual meeting systems, and constant notifications blur boundaries between professional and personal life, producing cognitive fatigue, decision exhaustion, reduced creativity, and worklife imbalance (Padmaavathy et al., 2026, p. 6). Similar mechanisms operate in educational settings, where dependence on AIassisted learning systems and recommendation algorithms may weaken attention regulation and independent problemsolving, thereby contributing to academic strain and student burnout (Padmaavathy et al., 2026, p. 16). These findings underscore that digital engagement intensity, design patterns, and algorithmic incentives jointly determine whether social media participation supports or undermines wellbeing. The concept of sustainable digital wellbeing provides an integrative lens for product decisions in prosocial platforms. Sustainable approaches aim to optimize human flourishing within AIenabled environments by balancing productive technology use with psychological health, recognizing that problems arise when engagementmaximizing designs compromise autonomy, digital literacy, and ethical safeguards (Padmaavathy et al., 2026, p. 6; Padmaavathy et al., 2026, p. 12). In practical product development for a prosocial social app, these insights justify replacing dark patterns and exploitative reinforcement strategies with features that support healthy emotional regulation, attention control, and genuine social connectedness, thereby directly targeting the burnout risks documented in current research (Padmaavathy et al., 2026, p. 6; Ronald, 2025, p. 6; Dadage, 2026, p. 11).

1.2 Regulatory and Governance Landscape

Regulatory and governance arrangements form a critical boundary condition for any attempt to embed ethical safeguards and intentional user controls into social media products oriented toward digital wellbeing. Existing AI governance initiatives in healthcare and education predominantly focus on data protection, cybersecurity, and algorithmic fairness, while rarely addressing psychological autonomy and emotional stability as explicit regulatory objectives (Sharma, 2025, p. 2). This technocratic emphasis creates a structural gap: the governance of AI is increasingly institutionalized, yet the governance of digital wellbeing as a public good remains largely undefined, leaving prosocial platforms without clear external benchmarks for cognitivehealth protection (Sharma, 2025, p. 4). Within AIenabled occupational health monitoring, privacy, informed consent, and data minimization are treated as foundational ethical requirements. Computational systems that analyse wearable sensor data, digital communication logs, and behavioural patterns can expose intimate dimensions of emotional wellbeing, social interaction, and cognitive functioning, which increases the risk of misuse, unauthorized access, and discrimination without stringent safeguards. Ethical governance in such environments demands transparent, revocable consent protocols, strict limits on data retention to what is necessary for legitimate health objectives, and algorithmic transparency that makes predictive processes explainable and auditable. These principles directly inform the design of social applications that monitor or nudge user behaviour: any burnoutrelated analytics or adaptive support features must be bounded by consent,

minimization, and transparency so that AI remains a supportive wellness mechanism rather than an intrusive control tool. Algorithmic bias and fairness concerns further constrain responsible design. In healthcare monitoring, biased training data and institutionspecific models can reproduce inequities across gender, race, age, and professional hierarchy, misclassifying stress levels or overlooking vulnerability in underrepresented groups (Sarkar & Kudapa, 2024, p. 18). Bias mitigation is framed as a moral obligation, requiring diverse data representation, fairnessaware model design, continuous auditing, and attention to accessibility so that digital mental health resources remain usable for staff irrespective of literacy, role, or cultural background. Parallel analyses of AI systems across recruitment, healthcare, education, and content moderation similarly document unequal outcomes arising from biased data or flawed design assumptions, with discriminatory results reinforcing social inequalities and eroding trust among marginalized communities (Ronald, 2025, p. 7). For a prosocial social app, such findings impose a requirement to treat fairness and inclusion as core wellbeing objectives, not peripheral technical issues, and to embed systematic bias monitoring into the product development lifecycle (Ronald, 2025, p. 7; Sarkar & Kudapa, 2024, p. 18). Empirical work on AI in digital marketing suggests that users' perceptions of fairness are often mediated by whether organizations implement clear accountability mechanisms; surveys of platform users indicate that transparency and complaint channels can materially affect trust in AI interventions on social media (Jadallah, 2025, p. 2). This evidence may suggest that governance instruments for digital wellbeing should explicitly require accountability processes (such as audits and redress pathways) in addition to technical fairness checks (Jadallah, 2025, p. 2). Broader policy frameworks articulate riskbased and rightsoriented controls that are directly relevant to social media governance. The EU Artificial Intelligence Act introduces a risktiered structure and explicitly prohibits manipulative systems that deploy subliminal techniques or exploit vulnerabilities to distort behaviour, although current exceptions leave gaps in the protection of individual identity and raise difficult enforcement questions regarding the boundaries of prohibited manipulation. Complementary platform regulations such as the Digital Services Act and Digital Markets Act require large platforms to conduct risk assessments that include threats to mental health and democracy, mandate annual evaluations of impacts on fundamental rights, and impose algorithmic transparency obligations related to content moderation, competition, and user rights (Papari et al., 2026, p. 12). These instruments collectively signal that dark patterns and addictive design mechanisms are not merely a matter of design ethics but fall under formal regulatory scrutiny when they threaten mental autonomy, cognitive liberty, and mental integrity. Governance perspectives on digital wellbeing emphasise that power, agency, and accountability must be reconceptualized for intelligent systems. Current national and international policies, including the EU AI Act, UNESCO's Recommendation on the Ethics of Artificial Intelligence, India's National Digital Health Mission, and National Education Policy 2020, advance AI adoption while offering limited safeguards for the emotional and cognitive effects of continuous digital engagement. In response, the Governance of Digital Wellbeing Framework proposes five pillars for ethical oversight across education and mental health platforms: ethical accountability, algorithmic transparency, human oversight, psychological safety, and inclusive policymaking (Sharma, 2025, p. 2). Translating these pillars into concrete policy imperatives yields requirements such as nationallevel commissions dedicated to digital wellbeing, mandatory AI impact assessments focusing on psychological autonomy and cognitive diversity, and national standards for algorithmic wellbeing metrics that move beyond conventional data protection indicators toward measures of attention sustainability, authentic emotional engagement, and autonomy preservation. For institutional actors, governance is operationalized through digital wellbeing charters, internal ethics committees for AI oversight, and professional training in algorithmic literacy for educators, clinicians, and administrators (Sharma, 2025, p. 12). These mechanisms reorient governance from narrow compliance to shared responsibility for experiential outcomes, aligning with research that calls for humancentered, wellbeingoriented AI design where transparency, explainability, user agency, and ethical principles guide system development and evaluation (Ronald, 2025, p. 7; Padmaavathy et al., 2026, p. 17). Evidence from analyses of FinTech and broader AI adoption similarly points to a governance shortfall: emerging technologies create pressure for new regulatory geographies and supervisory models that can respond to privacy, fairness, and accountability risks at scale (Shahid et al., 2025, p. 11). That literature appears to caution that without institutional adaptation, new oversight mechanisms, updated legal frameworks, and clearer standards for ethical use, technical fixes alone are unlikely to secure cognitive and emotional protections in digital platforms (Shahid et al., 2025, p. 11).

2 Theoretical Perspectives Informing MindfulFeed

2.1 Behavioral Economics and Choice Architecture

Behavioral economics provides a structured account of why interface details such as button placement, color schemes, wording, and navigation structures can substantially shape user decisions in digital environments. Minor visual and interactive elements alter the salience of options, influencing which actions users perceive as preferable or effortless, and thereby establishing core conceptual foundations for analysing deceptive or manipulative practices in online systems (T & R, 2026, p. 2; Gray et al., 2021, p. 12). Within consent banners and privacy dialogs, designers may employ larger or brightly colored acceptance buttons, preselected boxes, and hidden configuration links to prioritise agreement over refusal, turning neutral choice architecture into interface interference oriented dark patterns that systematically steer users toward tracking or data disclosure choices (Gray et al., 2021, p. 12). These salience manipulations exploit limited attentional resources and cognitive shortcuts, especially when users are multi tasking or processing information quickly, and can produce routinised, unreflective default behaviours even when explicit nudging elements are reduced (Gray et al., 2021, p. 12; Gray et al., 2021, p. 14). For MindfulFeed, these findings delineate specific design moves that must be excluded from consent, notification, and personalization flows if the product is to avoid replicating attention diversion and traffic light metaphor biases that make psychologically costly options less visible or harder to enact (Gray et al., 2021, p. 12). Dual process models and nudging theory further clarify the mechanisms that dark patterns employ. Kahneman's distinction between fast, intuitive System 1 and slower, deliberative System 2 processing shows that users under time pressure or uncertainty rely heavily on heuristics, which increases susceptibility to interface structures that exploit anchoring, loss aversion, default effects, and least effort principles (T & R, 2026, p. 2; P et al., 2026, p. 2). Thaler and Sunstein's notion of nudging conceptualises subtle changes in choice architecture that encourage particular behaviours without formally restricting options; these can be deployed for socially beneficial purposes or for organisational gains that sideline user welfare (T & R, 2026, p. 2). Empirical work on dark patterns in e commerce websites demonstrates how countdown timers, hidden costs, and misleading consent mechanisms can systematically increase conversion and data sharing, confirming that these architectures actively harness cognitive biases rather than merely presenting information (P et al., 2026, p. 2; T & R, 2026, p. 2). In the specific context of cookie banners and consent management, preselected options and obstructive flows have been shown to reduce privacy preserving choices and nudge users toward acceptance, thereby transforming a nominally free decision into one architecturally determined by interface configuration (P et al., 2026, p. 2; Gray et al., 2021, p. 3). Any behavioral blueprint for MindfulFeed must therefore treat defaults, framing, and friction levels in opt in and opt out pathways as primary levers for ethical design, ensuring symmetric effort for privacy friendly actions and avoiding guilt or loss framing in calls to action (P et al., 2026, p. 8; Gray et al., 2021, p. 12). Ethical analyses distinguish between dark patterns and ethical nudges by reference to organisational intent and autonomy outcomes. Dark patterns primarily manipulate users to maximise organisational metrics such as engagement, subscriptions, or sales, frequently undermining informed consent, transparency, and perceived control (T & R, 2026, p. 2; P et al., 2026, p. 2). In consent interfaces, obstruction, forced action, and interface interference strategies can render rejection cumbersome or degrade service quality for those who decline tracking, thereby placing disproportionate cognitive and experiential burdens on privacy oriented users (Gray et al., 2021, p. 3; Gray et al., 2021, p. 12). Research on "bright patterns" and educative nudges suggests limited practical feasibility in commercial settings, since many companies remain structurally incentivised to track users and to deploy architectures that favour data collection; even pro privacy nudges risk reproducing unreflective defaults and perceived lack of control when they rely solely on automatic steering rather than meaningful deliberation (Gray et al., 2021, p. 15). For a pro social app such as MindfulFeed, these constraints imply that ethical choice architecture must foreground user comprehension and reversible, low friction controls over behaviourally optimized funnels, integrating educative elements about data consequences without subordinating them to engagement maximisation (Gray et al., 2021, p. 15; Gray et al., 2021, p. 3). Aligning cancellation, consent withdrawal, and configuration pathways with principles of balanced salience, equal optical prominence for accept and reject options, and unambiguous consent satisfies both behavioural evidence on autonomy supportive design and legal expectations that withdrawing consent be as easy as giving it (Gray et al., 2021, p. 12; Gray et al., 2021, p. 3; P et al., 2026, p. 2). Recent work on AIGC mediated cultural production suggests an additional layer of complexity: when

algorithmic systems act as active co-creators of content and interaction, users' expectations and attentional habits are reshaped in ways that may interact with interface nudges, making it more likely that subtle salience shifts produce routinised acceptance in contexts where AI-generated companions or personalized feeds appear socially normative (Gong, 2026, p. 4). This implies MindfulFeed should consider not only superficial parity in button design, but also how AI-driven content dynamics influence perceived normality and thereby the effectiveness of consent architectures (Gong, 2026).

2.2 Psychological Needs and Motivation

Human motivational dynamics in AImediated environments are shaped by the interaction between fundamental psychological needs and platform design features that systematically address, and sometimes exploit, these needs. SelfDetermination Theory (SDT) identifies autonomy, competence, and relatedness as core requirements for sustained motivation and wellbeing. Digital platforms explicitly target these needs by configuring personalized experiences that allow individuals to regulate content choices, by embedding gamified feedback such as achievement systems and progress indicators to reinforce competence, and by enabling community formation and social networking that foster relatedness through persistent online interaction. Artificial intelligence intensifies this motivational alignment. Recommendation systems and conversational agents tailor content and interactions to individual preferences, thereby increasing perceived relevance and satisfaction. Personalized interfaces, predictive text, and adaptive notifications collectively reduce behavioural friction, making it easier for users to continue satisfying needs within the same environment. Over time, repeated satisfaction through loweffort interactions transforms intentional engagement into habitual routines, such as automatically checking smartphones after waking, opening social media during breaks, or browsing recommended videos before sleeping, even when users consciously recognise that their digital engagement has become excessive and difficult to regulate. For a prosocial app such as MindfulFeed, these dynamics have direct design implications. SDT explains both the attractiveness and the potential psychological risks of AImediated experiences. Excessive algorithmic personalization may gradually erode genuine autonomy by subtly shaping behavioural options through predictive recommendations, shifting users from active selection to guided patterns of engagement. In parallel, persuasive technology principles describe how reminders, social comparison, gamification, reward systems, and behavioural feedback are intentionally configured to influence attitudes and behaviours without overt coercion. AI systems learn which persuasive strategies are most effective for specific individuals, thereby increasing the potency of nudges that anchor users in ongoing interaction cycles. Uses and Gratifications Theory adds a complementary lens for product decisions. Individuals adopt and continue to use digital technologies to satisfy diverse psychological and social needs, including information seeking, entertainment, identity development, emotional regulation, social interaction, professional networking, and education. AI expands the breadth and intensity of gratifications by supplying highly personalized streams of entertainment, adaptive learning experiences, predicted consumer offerings, emotionally responsive conversational agents, and individualized wellness guidance. This diversification of gratifications strengthens habitual engagement and demonstrates that problematic use does not arise solely from darkpatterned interface design, but also from repeated reliance on digital environments to meet emotional and psychological needs that may otherwise remain unmet. Within MindfulFeed's development record, motivational theories therefore function as operational constraints on feature ideation and prototyping. SDT requires that autonomy support be authentic rather than superficial, so personalization controls, recommendation explanations, and adjustable notification regimes must be implemented to preserve selfdirected choice rather than to lock users into predefined paths. Competence reinforcement through feedback and progress indicators should be oriented toward meaningful skill building or reflective selfknowledge instead of raw engagement metrics, while relatedness features need to promote constructive social connection rather than purely comparative or competitive dynamics. Persuasive technology insights justify embedding goal tracking and behavioural feedback only when they are transparently presented as wellbeing tools rather than covert engagement maximizers. Evidence on habit formation and gratifications advises that design sprints within the Empathize and Define phases focus on mapping which psychological needs different personas attempt to satisfy through social media, and how AIdriven personalization currently shapes those efforts. Low and midfidelity prototypes should deliberately test alternative ways of meeting autonomy, competence, and relatedness needs with higher transparency and user control, while empirical usability evaluations must capture whether features intended as supportive begin to function as de facto dependency loops. Anchoring product decisions in SDT, Uses and Grat-

ifications, and Persuasive Technology Theory ensures that motivational mechanisms in MindfulFeed are harnessed for sustainable digital wellbeing rather than for intensified engagement (Padmaavathy et al., 2026, p. 8). Evidence from domains where AI surveillance and automated monitoring have been deployed suggests a parallel risk: systems that reduce human oversight and accelerate automated detection can increase public mistrust and raise fraught questions about data retention, secondary uses, and accountability (Badhan et al., 2024, p. 2). Such findings imply that MindfulFeed should not only provide user controls and transparent explanations, but also consider independent audit mechanisms and clear limits on data reuse to avoid replicating the kinds of trust deficits observed in other AIheavy contexts (Badhan et al., 2024, p. 2).

2.3 Algorithmic Personalization and Dark Patterns

Algorithmic personalization operates as a central mechanism through which dark patterns scale and adapt to individual users in contemporary digital environments. Personalized recommendation systems on social media, streaming services, e commerce platforms, and news aggregators use machine learning to tailor content to inferred preferences, thereby reducing information overload and increasing perceived relevance for many users. At the same time, these systems tend to surface emotionally charged and sensational content because such material drives higher engagement, which has been empirically associated with increased polarization, exposure to misinformation, and emotional exhaustion (Ronald, 2025, p. 5). This engagement maximization dynamic interacts directly with dark pattern design: when attention capture is the dominant objective, interface configurations that privilege certain actions, conceal alternatives, or obstruct exit paths become instrumentally valuable, reinforcing compulsive use and undermining cognitive well being. For MindfulFeed, these findings delineate a critical boundary condition for algorithmic features: recommendation logic must be constrained by well being criteria rather than pure engagement metrics, and interface structures around personalized content must avoid exploitative manipulation. Research on dark patterns identifies a range of mechanisms that are particularly potent when combined with personalization. Foundational taxonomies describe patterns such as trick questions, privacy zuckering, misdirection, disguised ads, roach motels, and sneaking, each defined by specific manipulative strategies (P et al., 2026, p. 2; Gray et al., 2021, p. 4). Subsequent work systematizes these into higher level categories including nagging, obstruction, sneaking, interface interference, and forced action, with explicit attention to how interfaces privilege certain choices or require users to perform unwanted actions to continue accessing functionality (Gray et al., 2021, p. 4). Empirical experiments demonstrate that dark pattern consent mechanisms can reduce privacy preserving decisions by measurable margins, and that the design of cookie banners strongly determines consent withdrawal rates, indicating that nominally free choices are often architecturally dictated (P et al., 2026, p. 2; Gray et al., 2021, p. 4). Within an AI personalized environment, these mechanisms can be targeted at specific user segments whose behavioural data indicate higher susceptibility, making manipulative designs harder to detect and resist. MindfulFeed's product development record must therefore treat any personalization of consent, settings, or content controls as ethically sensitive, ensuring that tailored interfaces do not exploit user specific vulnerabilities. Behavioural economics provides further insight into how algorithmic personalization amplifies dark pattern effects. Dual process theory situates dark patterns as leveraging fast, automatic System 1 processing while bypassing slower, deliberative System 2 oversight. Anchoring, loss aversion, default effects, and least effort principles explain why preselected options, countdown timers, and asymmetric opt in versus opt out flows reliably steer decisions (P et al., 2026, p. 2). When personalization learns which specific anchors or frames work best for an individual, these biases are engaged with increasing precision. Studies note that sophisticated personalization technologies can tailor deceptive interfaces to consumer characteristics, thereby intensifying manipulative practices and making dark patterns more difficult to detect and resist. For MindfulFeed, this evidence imposes a design constraint: adaptive algorithms may be used to reduce friction for autonomy supportive actions, but not to calibrate salience, defaults, or emotional framing in ways that systematically favour data disclosure or prolonged engagement. Ethical and regulatory analyses connect these technical dynamics to privacy and autonomy concerns. Contemporary platforms frequently deploy complex privacy policies, confusing consent flows, and defaults that encourage unintended data disclosure, limiting user understanding of data collection, processing, and sharing (T & R, 2026, p. 3). As personalization, targeted advertising, and recommendation systems become more sophisticated, researchers argue that such technologies can amplify manipulative practices by tailoring interfaces against user interests, thereby raising the need for stronger ethical guidelines and

regulatory frameworks that prioritise welfare alongside business objectives (T & R, 2026, p. 3; Ronald, 2025, p. 7). Legal scholarship on consent banners under the GDPR and ePrivacy Directive underscores that interface design is central to lawful processing: consent must be freely given, specific, informed, and unambiguous, and dark patterns that rely on obstruction, forced action, or interface interference are incompatible with these requirements (Gray et al., 2021, p. 4). MindfulFeed's algorithmic personalization strategy must therefore be aligned with data protection by design, avoiding both overt and emergent dark patterns, and subjecting personalized flows to systematic ethical and legal review. Evidence on digital well being oriented AI design reinforces this orientation. Human centred and well being oriented frameworks argue for systems that prioritize transparency, explainability, user agency, and ethical principles, with success measured in terms of human flourishing rather than engagement alone (Ronald, 2025, p. 5; Ronald, 2025, p. 7). In this context, personalization should be deployed to support psychological stability and autonomy, not to deepen dependency. For the Empathize and Define phases in MindfulFeed's development, this implies collecting user narratives about how recommendation systems and consent interfaces have previously contributed to fatigue or loss of control, and using these insights to specify requirements such as symmetric effort in opt out paths, plain language in personalized prompts, and explicit explanations of why particular content or choices are being surfaced (P et al., 2026, p. 2; T & R, 2026, p. 3; Gray et al., 2021, p. 4).

3 Methodological Rationale and Design Thinking Process

3.1 Selection of Stanford d.School Framework

Selection of the Stanford d.School design thinking framework is grounded in methodological choices that prioritise governance, ethical accountability, and psychological outcomes in AImediated environments. Research on digital wellbeing governance adopts qualitative, multilayered designs that integrate policy analysis, theoretical synthesis, and case illustration to address complex sociotechnical questions, explicitly favouring depth, contextual understanding, and interdisciplinary integration over narrow technical evaluation (Sharma, 2025, p. 3; Sharma, 2025, p. 7). This orientation towards interpretivist, governancefocused inquiry aligns closely with design thinking's emphasis on understanding lived experience and experiential outcomes before proposing solutions. The Governance of Digital Wellbeing Framework operationalises this interpretive stance through four analytical dimensions: accountability, transparency, oversight, and wellbeing outcomes, all examined via thematic content analysis of policies and case evidence (Sharma, 2025, p. 8). These dimensions map directly onto the kinds of questions that each phase of the Stanford d.School process can structure in a product development record for MindfulFeed. For instance, accountability and human oversight can be systematically explored in early empathic research and problem definition, while transparency and psychological safety can be operationalised during ideation and prototyping of features that avoid dark patterns and respect cognitive autonomy. Evidence indicates that contemporary AI deployments in education and mental health are governed primarily through data protection, cybersecurity, and algorithmic fairness, leaving emotional stability, cognitive autonomy, and social connection comparatively underspecified in policy instruments (Sharma, 2025, p. 3; Sharma, 2025, p. 4). In parallel, studies of dark patterns document extensive use of manipulative interface strategies, strong behavioural impacts on consent and decision quality, and limited consumer awareness of such practices (P et al., 2026, p. 2; Mavi, 2025, p. 2; Ukgoda, 2024, p. 3). This combination of governance blind spots and manipulative interface prevalence creates a design problem that is as much ethical and experiential as it is technical. A phased design thinking framework provides a structured way to translate these abstract governance deficits and interaction harms into concrete product requirements, prototypes, and test protocols that can be traced through an evidencebased development record. The methodological approach used in digital wellbeing governance research relies on iterative coding, triangulation of policy, academic, and case data, and explicit procedures to ensure credibility, transferability, and dependability (Sharma, 2025, p. 8). Similarly, systematic reviews of AIbased burnout interventions employ staged processes of identification, screening, eligibility assessment, and inclusion, supported by structured coding frameworks, thematic clustering, and mixed methods appraisal tools to sustain analytical integrity (Sarkar & Kudapa, 2024, p. 19). Selecting the Stanford d.School framework allows MindfulFeed's development documentation to emulate this procedural clarity: each phase can be associated with defined activities, data sources, and analytic steps, such as empathy interviews, cognitive walkthroughs, prototype evaluations, and

usability diaries, all logged as traceable artefacts analogous to coded documents in policy analysis or screened studies in systematic reviews. Research on dark patterns highlights mechanistic clarity and nonoverlapping taxonomies as central design principles for classifying manipulative strategies, reflecting a need for exhaustive yet structured categorisation in interface assessment (P et al., 2026, p. 2; Mavi, 2025, p. 2). Cognitive walkthrough studies on social platforms and mindfulness apps further show that expert evaluators can systematically identify combinations of dark patterns across account creation, content engagement, and termination flows when guided by clear methodological protocols (Ukgoda, 2024, p. 3; Özkürkçü & Doğanay, 2025, p. 4). Embedding these taxonomic insights within the Define and Ideate phases of the d.School process ensures that MindfulFeed's problem statements and concept generation explicitly reference documented dark pattern categories, rather than relying on informal intuitions, and that subsequent prototypes can be evaluated against recognised harms such as obstruction, sneaking, confirm shaming, and roach motel behaviours. Finally, governance research on digital wellbeing insists that algorithmic systems be accountable to the moral conditions of human life, positioning attention, emotion, and cognition as contested frontiers that require ethical reorientation in power, agency, and accountability (Sharma, 2025, p. 3; Sharma, 2025, p. 4). A design thinking framework that begins with empathic investigation and iteratively cycles through prototyping and testing offers a practical methodology to realise this reorientation within a single prosocial app: each phase can explicitly incorporate governance pillars such as ethical accountability, transparency, human oversight, psychological safety, and inclusive stakeholder input, thereby aligning productlevel decisions with broader policy ambitions for humancentred AI systems (Sharma, 2025, p. 4; Judijanto, 2026, p. 6).

3.2 Comparison with Alternative Design Frameworks

Evaluating the suitability of the selected design thinking framework requires comparison with alternative methodological approaches that have been applied to digital wellbeing, dark patterns, and AI governance. Qualitative, multilayered policy analysis designs integrate policy review, conceptual synthesis, and comparative case examination to construct governance frameworks such as the Governance of Digital Wellbeing Framework. These approaches privilege contextual depth and interpretive analysis of accountability, transparency, oversight, and wellbeing outcomes rather than structured product development activities (Sharma, 2025, p. 7; Sharma, 2025, p. 8). In contrast, the Stanford d.School process used in MindfulFeed explicitly organises activities around empathic research, problem definition, ideation, prototyping, and testing, which more directly supports the creation and iterative evaluation of concrete interface artefacts, consent flows, and darkpattern countermeasures within a product development record. Systematic review methodologies provide another point of comparison. PRISMAguided reviews of AIbased burnout interventions employ staged identification, screening, eligibility, and inclusion procedures, coupled with structured coding frameworks, thematic clustering, and formal quality appraisal tools such as the Mixed Methods Appraisal Tool. These designs achieve methodological transparency, replicability, and quantitative documentation of trends in intervention effectiveness and predictive accuracy, but they focus primarily on synthesising existing empirical studies rather than generating and testing new interface solutions. For MindfulFeed, such systematic review procedures are informative for evidence gathering and risk mapping, yet they lack the phasebased structure for producing and iterating low, mid, and highfidelity prototypes that is central to the product brief (Sarkar & Kudapa, 2024, p. 19). Within human–computer interaction and user experience research, expertled cognitive walkthroughs represent a distinct methodological paradigm. Cognitive walkthrough studies on large social platforms conduct taskbased evaluations across content engagement, account creation, and termination flows to identify combinations of dark patterns and analyse how these combinations affect user experience and engagement. Sampling of multiple platforms and pilot testing of procedures ensure coverage of diverse interaction modalities and refinement of feature categorisation and evaluator guidelines. This approach yields detailed diagnostic insight into existing interface harms and dark pattern configurations, yet it is not inherently organised around participatory ideation or prototype generation for alternative, autonomysupportive designs (Ukgoda, 2024, p. 2; Ukgoda, 2024, p. 3). Integrating cognitive walkthrough techniques into the Test phase of the d.School framework therefore enhances evaluative rigour while preserving the overall innovationoriented trajectory of the product development record. Multimethod field studies of mindfulness applications illustrate another alternative framework. Over several weeks, combined heuristic evaluation, surveys, diaries, and user interviews are used to examine dark patterns and user perceptions in popular apps. Heuristic evaluation

applies an established dark pattern taxonomy, including work by Brignull and subsequent projects, while surveys and diaries provide longitudinal data on engagement, discontinuation, and user attitudes. This multistage design mirrors certain aspects of empathic and evaluative research, but it is structured primarily for understanding existing apps rather than for guiding the stepwise construction of a new prosocial platform with explicit regulatory and theoretical anchors (Özkürkçü & Doğanay, 2025, p. 2; Özkürkçü & Doğanay, 2025, p. 4). For MindfulFeed, diary and interview components can be embedded within Empathize and Test phases, yet the overarching design thinking framework remains more suitable for mapping desirability, feasibility, and ethical constraints into iterative prototypes. Dark pattern research itself introduces taxonomic and automated detection frameworks that could, in principle, be treated as design methodologies. Comprehensive taxonomies emphasise mechanistic clarity, nonoverlap, and exhaustiveness, organising dozens of patterns into a small number of toplevel categories. Automated detection pipelines such as DarkScan combine static DOM analysis, text classification, visual detection, and dynamic behavioural tracing to identify manipulative strategies and to quantify behavioural and psychological impacts on decision quality, trust, and wellbeing. These frameworks are optimised for classification, measurement, and policy recommendation rather than for participatory codesign of alternative interaction models. When compared with the d.School structure, taxonomic and detection approaches function as analytical backbones that inform problem framing, constraint identification, and success metrics, but they do not replace the need for phases dedicated to ideation, prototyping, and iterative user testing (P et al., 2026, p. 2). Finally, interaction criticism and legalethical analyses of consent banners embody interpretive designresearch frameworks focused on uncovering emergent dark patterns and evaluating compliance with consent requirements. Such work reveals previously uncategorised strategies like consent walls and reduced service, discusses bright patterns and educative nudges, and interrogates the burdens placed on end users by current consent regimes. This perspective highlights power asymmetries and regulatory constraints, yet it stops short of prescribing a procedural pathway for generating and validating alternative interfaces at scale. Within MindfulFeed's development record, interaction criticism is incorporated as an evaluative lens in later phases, while the d.School framework provides the operational sequence through which evidence, ethics, and user narratives are translated into concrete design decisions, prototypes, and documented test outcomes (Gray et al., 2021, p. 15).

4 Empathize Phase: Understanding Digital Burnout and User Needs

4.1 Qualitative Exploration of Digital Burnout

Empathic investigation of digital burnout within MindfulFeed's product development record requires qualitative methods capable of capturing how manipulative design strategies and algorithmic environments shape emotional stability, cognitive autonomy, and social connectedness. Governance research on digital wellbeing adopts thematic content analysis guided by four dimensions, namely accountability, transparency, oversight, and wellbeing outcomes, using interpretive coding procedures to link administrative and technological design with experiential consequences for users' cognition and emotion (Sharma, 2025, p. 7; Sharma, 2025, p. 8). This analytical structure directly informs the qualitative exploration strategy for MindfulFeed: semistructured data collection must explicitly probe how users attribute responsibility for harmful engagement loops, how visible and explainable recommendation and notification systems appear, what mechanisms they experience as human oversight or redress, and which emotional or social outcomes they associate with prolonged platform use. Dark pattern research in social media and educational technologies underscores that manipulative interface combinations can exacerbate distraction during learning and exposure to harmful content, lowering student achievement and undermining the educational potential of platforms that might otherwise support wellbeing (Ukgoda, 2024, p. 3). Cognitive walkthrough studies across major social media services such as YouTube, LinkedIn, WhatsApp, and Telegram show that UI/UX experts, when guided by structured tasks, can systematically identify combinations of dark patterns in content creation, engagement, account creation, and account termination flows, revealing how these patterns interact to shape user behaviour and overall experience (Ukgoda, 2024, p. 2; Ukgoda, 2024, p. 3). These findings justify including expert walkthroughs alongside user interviews in the Empathize phase, with each walkthrough documented as a qualitative artefact that maps specific pattern clusters to hypoth-

esised burnout drivers such as heightened cognitive load, difficulty disengaging, or obstructed account exit. Multimethod fieldwork on mindfulness applications provides an additional template for exploring burnoutrelated experiences in app ecosystems that explicitly address psychological health. Over nine weeks, heuristic evaluation, surveys, diaries, and user interviews were combined to examine how dark patterns in Calm, Headspace, and Insight Timer influence user perceptions and engagement, with ethical clearance obtained in advance and standardized actions such as signup and deletion used to identify manipulative designs. Diary studies over five days allowed participants to use assigned apps flexibly while logging usage, activities, reasons for nonusage, and perceptions of trust and engagement, thereby supporting familiarity with the app and sensitivity to subtle manipulative cues (Özkürkçü & Doğanay, 2025, p. 4). Subsequent interviews used a threepart structure that first elicited general insights on mindfulness and design features, then presented expert review videos of dark patterns to prompt reflection, and finally employed guided application walkthroughs where users shared screens and discussed account deletion or unsubscribing processes in real time. Data analysis followed an inductive content analysis procedure, with repeated reading of transcripts, coding of similar information, and theme construction around perception, motivation, behaviour, and dark patterns, revealing user types (skeptics, interested, experienced) and documenting correlations between dark pattern exposure and negative behaviours such as fleeing the application (Özkürkçü & Doğanay, 2025, p. 5). Adapting this structure to MindfulFeed, the qualitative exploration of digital burnout will similarly combine diaries of social media use, videobased expert walkthroughs of problematic flows, and semistructured interviews focused on attentional strain, emotional exhaustion, and perceived loss of control. Normative analysis of algorithmic addictive designs directed at older adults identifies aADs as systems that use predictive analytics and reinforcement loops to subtly shape behaviour, eroding selfdirected decision making and impairing autonomy in ways comparable to coercive contracts that are invalid in law. Excessive technology use is linked to mental health issues, reduced productivity, and increased risks of substance abuse, yet regulatory models often treat aADs as isolated design flaws rather than structural market failures, suggesting that qualitative inquiries must explicitly address how users understand their own responsibility versus systemic manipulation in burnout trajectories. Behaviourally informed risk disclosure concepts such as concise labels indicating addiction risk scores, typical time commitments, and available disengagement tools provide a lens for interview questions about whether existing platforms communicate engagement risks adequately and how such labels might alter subjective experiences of burnout (Yao, 2025, p. 4). Culturally responsive regulation perspectives further emphasize that autonomy, care, and responsibility are interpreted differently across contexts, which argues for sampling diverse user groups and exploring how intergenerational expectations, social isolation, or rural marginalization intersect with digital burnout narratives (Yao, 2025, p. 5). Integrating these governance and dark pattern insights, MindfulFeed's qualitative exploration will generate empathy maps, coded themes, and illustrative quotes that document how manipulative architectures, opaque recommendations, and inadequate disclosure contribute to burnout, thereby grounding subsequent Define phase problem statements in systematically analysed experiential evidence (P et al., 2026, p. 2; T & R, 2026, p. 2; Sharma, 2025, p. 7; Özkürkçü & Doğanay, 2025, p. 5). Recent policy and media analyses further suggest that transparency obligations and explicit content labelling may alter how users make sense of algorithmic curation; in particular, mandatory disclosures and clearer markers for algorithmic personalization are likely to affect perceptions of accountability and editorial responsibility (D, 2025, p. 9). Such provisions, which aim to make algorithmic influence more visible and to require human oversight in editorial workflows, will be considered as prompts in interviews and as contextual variables when interpreting diaries and walkthrough artefacts (D, 2025, p. 9).

4.2 User Interview Profiles and Triggers

Interview sampling for MindfulFeed's empathic work can operationalise profiles that mirror established qualitative studies on youth social media use and mindfulness apps. A purposive sampling strategy is appropriate for selecting information rich participants capable of articulating detailed experiences of digital fatigue, dark patterns, and emotional distress, as purposive sampling is explicitly recommended for qualitative designs seeking depth over breadth in youth social media contexts. Target groups can include students and young adults in dense educational settings similar to Bambili, where social media is an unavoidable component of daily life and interaction, ensuring diversity of experiences in terms of academic use, entertainment, communication, and psychological impacts such as cyberbullying and misinformation. Semi structured interviews provide a flexible yet focused format for exploring burnout

triggers. Interview guides should use open ended questions that invite narratives about motivations, routines, and challenges in social media use, while enabling probing into sensitive topics like emotional strain, distraction, and manipulative experiences, paralleling procedures shown to be effective for discussing cyberbullying, online scams, and distress in conversational settings. Each interview can be planned for approximately 15 to 20 minutes to maintain concentration and reduce respondent fatigue, with audio recording used for accurate transcription and subsequent thematic analysis, following manual coding practices that keep researchers closely connected to emerging patterns in subjective experience (Peguy & Precious, 2025, p. 5). Profile construction within the MindfulFeed record can draw directly on mixed methods findings from mindfulness application research. Users of mental health and mindfulness apps have been categorised into skeptics, interested, and experienced types, differentiated by their intrinsic and extrinsic motivations and associated behaviours. Skeptics predominantly exhibit extrinsic motivations and may be more sensitive to manipulative strategies, while experienced users show primarily intrinsic motivations and persist despite encountering dark patterns. Interested users, with mixed motivations, often report confusion and a need for guidance, making them particularly vulnerable to complex, rigid, or pressured interaction flows. These three motivational profiles can be repurposed for MindfulFeed's interviews, with targeted prompts and scenarios designed for each type to capture distinct burnout triggers and reactions to persuasive or manipulative design. Diary informed interview structures further refine trigger identification. In the referenced mindfulness study, participants first completed a five day diary capturing app usage, activities, experiences, and reasons for non usage, fostering familiarity and sensitivity to design nuances. Subsequent interviews used a three part format: initial discussion of general app usage and dark patterns, viewing of expert review videos to elicit reflections on negative impacts, and guided walkthroughs where participants shared screens and demonstrated account deletion or unsubscribing processes. Adopting this sequence for MindfulFeed allows interviews to surface triggers at multiple levels: narrative reflections on burnout, reactions to observed dark patterns, and real time commentary on navigation difficulties, obstructions, or confusing flows in termination and configuration tasks. Analytical procedures for these interviews should follow inductive content analysis. Semi structured interviews are transcribed, read repeatedly to identify similar and relevant information, and coded into categories that then form themes such as perception, motivation, behaviour, and dark patterns, as implemented in mindfulness app research (Özkürkçü & Doğanay, 2025, p. 5). Themes can then be mapped to specific burnout triggers, including cognitive overload from notifications, emotional strain linked to rigid progress streaks, or frustration arising from short trial periods that prevent habit formation and understanding of benefits; in prior work, users reported that 7 to 14 day trials were insufficient relative to habit formation estimates, leading to confusion and disengagement. Recognising dual constraints, where features like streaks or notifications are beneficial for some users but induce stress or guilt for others, is essential for interpreting heterogeneous trigger responses and for identifying design elements that must be configurable rather than universal (Özkürkçü & Doğanay, 2025, p. 8). To maintain methodological trustworthiness, interview data for MindfulFeed should be triangulated with other artefacts such as diaries and expert walkthrough observations, echoing multi method approaches that combine heuristic evaluation, survey, diary, and interviews to understand dark patterns' impact on perception and engagement (Özkürkçü & Doğanay, 2025, p. 4). Credibility, transferability, and dependability can be enhanced through detailed contextual descriptions of participant demographics, replicable interview guides, and transparent coding schemes, in line with governance oriented thematic content analysis that emphasises accountability, transparency, oversight, and well being outcomes in AI mediated environments (Sharma, 2025, p. 8).

5 Define Phase: Framing the MindfulFeed Design Challenge

5.1 Synthesis of Empathize Insights into Design Criteria

Empathic work on digital wellbeing and dark patterns provides concrete dimensions that can be translated into design criteria for MindfulFeed. The Governance of Digital Wellbeing research operationalises experiential concerns through four analytical dimensions: accountability, transparency, oversight, and wellbeing outcomes, each explicitly focused on emotional stability, cognitive autonomy, and social connectedness in AImediated environments (Sharma, 2025, p. 3; Sharma, 2025, p. 8). These dimensions become toplevel requirement categories in the Define phase. Interface and feature concepts must be specified in ways that allow later traceability: for every planned flow or control, the product

record should document which accountability mechanism is visible to users, how transparency about AI operations or data use is provided, which human oversight or feedback channel is available, and which wellbeing outcomes the feature is intended to protect. Methodological procedures used in governance studies and dark pattern detection further shape how empathize insights are converted into actionable criteria. Qualitative policy research emphasises multisource triangulation, detailed contextual descriptions, and transparent coding to ensure credibility, transferability, and dependability (Sharma, 2025, p. 8). Dark pattern work likewise stresses mechanistic clarity, nonoverlapping taxonomies, and exhaustive categorisation to support reliable identification and impact assessment (P et al., 2026, p. 2; Ukgoda, 2024, p. 3). In the product development record, these principles imply that user narratives, diary entries, and expert walkthrough observations must be coded against a clearly defined dark pattern taxonomy and the four GDW dimensions, yielding a matrix of "pain points by mechanism" that directly informs design criteria. For instance, if cognitive walkthroughs reveal obstruction and interface interference during account termination flows on major platforms, this is recorded as a specific problem mechanism to be reversed through symmetric, lowfriction exit designs (Ukgoda, 2024, p. 3; P et al., 2026, p. 2). Empathize findings from mindfulness app research add granular detail on how manipulative designs translate into negative user experience patterns. In a nineweek multimethod study, heuristic evaluation, diaries, surveys, and semistructured interviews documented how Brignulltype dark patterns such as bait and switch, confirm shaming, disguised ads, forced continuity, friend spam, and hidden cost appear within Calm, Headspace, and Insight Timer, and how users respond behaviourally and emotionally to these techniques (Özkürkçü & Doğanay, 2025, p. 2). Inductive content analysis organised interview data into themes of perception, motivation, behaviour, and dark patterns, and identified user types (skeptics, interested, experienced) with distinct sensitivities and motivations (Özkürkçü & Doğanay, 2025, p. 5). Translating these insights into MindfulFeed's Define phase requires explicit patterntocriterion mapping. Each observed pattern category becomes a banned mechanism, and each theme becomes a design objective: avoid guiltinducing copy for "confirm shaming", avoid hidden paywalls or short trial periods that undermine understanding, and ensure that any continuity or reminders are presented as configurable support tools rather than enforced obligations. Diary and guided walkthrough structures used in the mindfulness study also suggest practical execution details for MindfulFeed. Participants kept fiveday diaries capturing usage, activities, reasons for nonusage, and trust and engagement perceptions, then participated in interviews that combined general reflection, viewing of expert dark pattern reviews, and live screensharing for account deletion tasks (Özkürkçü & Doğanay, 2025, p. 4). For the Define phase, the product record should synthesise these types of artefacts into empathy maps and trigger lists, indicating which interface moments most frequently correlate with cognitive overload, emotional strain, or perceived loss of control. Each trigger is then formalised as a design criterion, such as limiting notification volume, eliminating countdown timers tied to mental health content, or ensuring that account deletion and data export can be completed in a small, documented number of steps. Governance analysis of digital wellbeing underscores that AI systems can erode agency by quantifying learning patterns and mental states into optimisable data streams, exposing individuals to surveillance, behavioural nudging, and psychological conditioning (Sharma, 2025, p. 3). Combined with dark pattern research showing measurable impacts on decision quality, trust, and psychological wellbeing, and demonstrating how consent interfaces can reduce privacypreserving choices by significant margins (P et al., 2026, p. 2), empathize insights must be synthesized into criteria that explicitly guard against agency erosion. This involves requiring plain language in calls to action, enforcing symmetric complexity for optin and optout journeys, banning consent walls and reducedservice strategies that tie access to data disclosure, and mandating that AIdriven personalisation be explainable and adjustable by users (Gray et al., 2021, p. 15; P et al., 2026, p. 8). Design criteria framed in this way create a direct bridge between users' burnout narratives and concrete interface and feature constraints, providing a structured problem definition that subsequent Ideate, Prototype, and Test activities can operationalise and evaluate. Evidence from related empirical work on shortform video consumption suggests that very high exposure can be associated with reduced attention spans and frequent, repeated app access, an outcome that may parallel the cognitive overload seen with manipulative interface patterns; such findings imply we should treat highfrequency content loops and reward scheduling as risk vectors when specifying wellbeing constraints (Dhillon et al., 2025, p. 10).

5.2 Primary Persona Development

Primary persona construction for MindfulFeed must be anchored in empirically derived motivational and behavioural profiles rather than hypothetical user archetypes. Mixed methods research on mindfulness applications identifies three distinct user types, differentiated by the nature of their motivations and their interaction with dark patterns: skeptics, interested users, and experienced users. Skeptics exhibit predominantly extrinsic motivations, display higher sensitivity to manipulative strategies, and are more likely to disengage or "flee" when confronted with opaque flows or pressured calls to action. Interested users show a blend of intrinsic and extrinsic motivations, frequently reporting confusion and a need for guidance in navigating complex app features. Experienced users are primarily intrinsically motivated, maintain more regular usage, and are more inclined to purchase applications they perceive as beneficial and use frequently. Defining a primary persona for MindfulFeed requires selecting the profile whose vulnerabilities and needs align most closely with the project brief's emphasis on counteracting dark patterns and digital burnout. Findings indicate that the presence of manipulative designs correlates with negative behaviours, particularly abandonment, and that motivational composition shapes how different users interpret and tolerate these experiences (Özkürkçü & Doğanay, 2025, p. 5). For an evidence based product record focused on ethical, autonomy supporting interaction, the interested user type provides the most critical baseline. This profile is simultaneously attracted to perceived benefits and susceptible to confusion, making it an appropriate focal point for design criteria that must reduce cognitive load, clarify flows, and avoid exploitative reinforcement. Semi structured interviews in mindfulness studies demonstrate that such users can articulate nuanced perceptions of design features and dark patterns when prompted with examples and guided walkthroughs, offering rich qualitative material for persona detail (Özkürkçü & Doğanay, 2025, p. 2; Özkürkçü & Doğanay, 2025, p. 5). Operationalising this primary persona requires systematic integration of diary entries, interview codes, and guided walkthrough observations. In the referenced research, a five day diary phase captured usage, activities, experiences, and reasons for non usage, fostering familiarity and sensitivity to subtle manipulative cues in Calm, Headspace, and Insight Timer (7173856 Özkürkçü & Doğanay, 2025, p. 4). Subsequent interviews followed a three part structure: open discussion of mindfulness and app usage, exposure to expert dark pattern review videos, and live screen sharing to examine routine tasks and account deletion or unsubscribing processes (Özkürkçü & Doğanay, 2025, p. 2; Özkürkçü & Doğanay, 2025, p. 4). Content analysis grouped transcribed data into codes and themes under perception, motivation, behaviour, and dark patterns, revealing how interested users interpret design elements, experience stress, and respond to pressured flows (7173856 Özkürkçü & Doğanay, 2025, p. 5). For MindfulFeed, persona attributes should be derived through equivalent coding, with fields such as typical usage context, motivational drivers, trust thresholds, burnout triggers, and reactions to obstructive or guilt framed interfaces populated directly from these thematic clusters. Trustworthiness requirements in governance oriented qualitative research further constrain persona development. Thematic content analysis in AI governance work emphasises credibility through data triangulation, transferability through detailed contextual descriptions, and dependability via transparent coding procedures (Sharma, 2025, p. 8). Applying these criteria, the primary persona must be documented with explicit references to source artefacts: diary excerpts demonstrating feelings of anxiety or irritation when access is restricted, interview segments describing difficulty identifying manipulative designs due to "dark pattern blindness," and walkthrough observations revealing confusion or frustration in account management tasks (7173856 Özkürkçü & Doğanay, 2025, p. 5). Research notes that repeated exposure can normalise dark patterns, leading users to justify or accept them despite residual discomfort, which is a critical experiential dimension that should appear in the persona's attitudinal profile and burnout risk assessment (7173856 Özkürkçü & Doğanay, 2025, p. 5). Dark pattern detection and governance studies also inform the behavioural envelope within which the primary persona operates. Cognitive walkthroughs on YouTube, LinkedIn, WhatsApp, and Telegram show that experts, when guided by structured tasks, can identify combinations of dark patterns across content engagement and account creation or termination, and that these combinations shape user experience and interaction difficulty (Ukgoda, 2024, p. 3). Comprehensive taxonomies and automated detection pipelines such as DarkScan document measurable impacts of dark pattern exposure on decision quality, platform trust, and psychological well being, and recommend symmetric design principles ensuring that opting out, cancelling, or exercising privacy preferences is no more complex than the corresponding opt in flows (P et al., 2026, p. 2; P et al., 2026, p. 8). MindfulFeed's primary persona therefore must explicitly encode expectations for symmetric pathways, plain language, and low friction autonomy safeguards, treating any

deviation from these expectations as a probable burnout or disengagement trigger. Finally, governance analysis frames attention, emotion, and cognition as contested frontiers in AI mediated environments, with digital well being defined as a problem of accountability, transparency, oversight, and well being outcomes (7808872 Sharma, 2025, p. 8). The primary persona for MindfulFeed should thus be structured around these four dimensions, specifying how the interested user type perceives responsibility for harmful engagement loops, understands or fails to understand algorithmic operations, experiences available oversight or complaint mechanisms, and links interface encounters to emotional stability and social connectedness. Anchoring persona development in this multi dimensional, empirically coded profile ensures that subsequent ideation and prototyping activities remain aligned with documented user needs and governance principles rather than speculative assumptions (7808872 Sharma, 2025, p. 8; P et al., 2026, p. 2; Ukgoda, 2024, p. 3). This body of work also signals that digitalisation requires coordinated technological, legal and social measures and that algorithmic audit and transparency are often necessary to prevent new forms of discrimination or privacy harm (Zhyhalkin & Chupryna, 2025, p. 2). Examples from national systems further suggest that decentralised data exchange layers and immutable logging can increase traceability and accountability, which complements the requirement for explicit traceable artefacts in persona records (Zhyhalkin & Chupryna, 2025, p. 2).

5.3 Strategic HowMightWe Questions

Strategic HowMightWe questions in the product development record must translate documented harms and governance concerns into open, solutionoriented prompts that remain tightly anchored in empirical findings on dark patterns, cognitive bias, and digital wellbeing governance. Dark pattern taxonomies grounded in mechanistic clarity, nonoverlap, and exhaustiveness show that manipulative strategies such as sneaking, obstruction, misdirection, forced action, confirm shaming, roach motels, and drip pricing deliberately exploit cognitive tendencies including default bias, scarcity bias, framing effects, and loss aversion to steer users into decisions they did not intend to make (Mavi, 2025, p. 2; P et al., 2026, p. 2). Within the Define phase, strategic questions must therefore ask how interface flows and consent dialogs can be architected so that these mechanisms are structurally impossible rather than merely discouraged, while still satisfying legal requirements that consent be freely given, specific, informed, unambiguous, and revocable under the GDPR and ePrivacy regimes (Gray et al., 2021, p. 4; P et al., 2026, p. 2). Evidence indicates that darkpattern consent mechanisms reduce privacypreserving choices by up to 23 percentage points, and that cookie banner design significantly determines consent withdrawal rates, demonstrating that apparent freedom of choice can be overridden by interface configuration (P et al., 2026, p. 2). A critical HowMightWe prompt for MindfulFeed is therefore: How might consent and privacy flows be structured so that opting out, withdrawing consent, or limiting tracking is at least as salient and effortless as opting in, eliminating asymmetric effort and routinised conditioning effects identified in experimental work on consent dialogs (P et al., 2026, p. 2; Gray et al., 2021, p. 14)? Complementary legal scholarship frames tracking walls and reduced service designs as unlawful or legally actionable, yet notes that such patterns continue to appear because current consent systems place substantial decision burdens on users in complex, cognitively demanding contexts (Gray et al., 2021, p. 4; Gray et al., 2021, p. 15). This motivates further questions about reducing cognitive load: How might MindfulFeed minimise the informational and attentional burden imposed at consent points, while still conveying meaningful data consequences through approaches such as "data visceralization" that connect users' feelings with their data in tangible, emotionally appreciable ways (Gray et al., 2021, p. 15)? Governance analysis conceptualises digital wellbeing as a problem of accountability, transparency, oversight, and wellbeing outcomes, highlighting how AI systems in education and mental health can erode human agency through surveillance, behavioural nudging, and psychological conditioning when learning patterns and mental states are optimised as data streams. Strategic HowMightWe questions must therefore address the governance frontier directly: How might interface and AIdriven features in MindfulFeed make accountability for algorithmic decisions visible to end users, clarify which actors bear responsibility for psychological outcomes, and provide oversight channels that permit feedback and correction when recommendation or notification behaviours contribute to cognitive strain or emotional distress (Sharma, 2025, p. 3; Sharma, 2025, p. 8)? Because attention, emotion, and cognition are described as contested frontiers analogous to "new oil", and because governance aims to function both as shield and compass for human dignity, a further question arises: How might MindfulFeed embed transparent explanations of algorithmic operations affecting feed ranking, prompts, and behavioural feedback so that users can understand and contest potential

nudging or conditioning (Sharma, 2025, p. 3)? Awareness research shows that while dark patterns are widely visible on digital platforms, most users fail to recognise them as intentional manipulation, often misclassifying them as mere bad design and demonstrating particularly low comprehension of datadriven dark patterns such as cookie consent manipulation (Mavi, 2025, p. 2). This calls for HowMightWe questions that combine design and education: How might MindfulFeed surface manipulative design risks in a way that increases user literacy about dark patterns without reproducing unreflective defaults or perceived lack of control associated with some "bright patterns" and proprivacy nudges (Gray et al., 2021, p. 15; Mavi, 2025, p. 2)? Since enduser centric consent regimes place substantial and possibly unwarranted pressure on individuals to make fully informed choices under uncertainty, another strategic question is: How might responsibility for ethical data practices be redistributed within MindfulFeed's ecosystem, for example by embedding default safeguards and constraint mechanisms at the system level, so that user choices operate within a protective envelope instead of being the sole barrier to exploitation (Gray et al., 2021, p. 4; Sharma, 2025, p. 3)? Finally, comprehensive taxonomies and automated detection pipelines such as DarkScan demonstrate that darkpattern exposure measurably degrades decision quality, platform trust, and psychological wellbeing, and they provide evidencebased design and policy recommendations rooted in regulatory frameworks (P et al., 2026, p. 2). Strategic HowMightWe questions must integrate these diagnostics into the development record, asking: How might MindfulFeed use taxonomic clarity and automated checks as internal design constraints to ensure that no pattern category from established lists is instantiated in any prototype flow, and how might behavioural and trust metrics from usability testing be monitored as primary success indicators rather than engagement or conversion (P et al., 2026, p. 2; Mavi, 2025, p. 2)? Framing HowMightWe prompts in this empirically grounded way keeps subsequent ideation tethered to demonstrated harms, governance dimensions, and legal thresholds, providing a structured bridge from burnout narratives and darkpattern evidence to actionable, autonomysupportive design requirements. Empirical work on platform amplification and early detection suggests useful operational details for monitoring and intervention: algorithms can create distinct promotion dynamics across platforms and early-warning models may detect escalation windows of a few hours, which appears to matter for when automated checks should fire and when human review is required (Prabowo et al., 2025, p. 11). That same research reports that culturally informed interventions markedly reduce escalation compared with generic fact-checking, implying that automated detection should be paired with response protocols that are sensitive to cultural and linguistic context rather than relying solely on blunt suppression (Prabowo et al., 2025, p. 11).

6 Ideate Phase: Concept Generation for ProSocial Social Media

6.1 Ideation Methods and Creative Exploration

Ideation for MindfulFeed must operationalise ethical, governance, and dark pattern evidence into concrete concept generation procedures that can later be prototyped and evaluated. Qualitative governance work on digital wellbeing demonstrates the usefulness of multilayered analysis and thematic content coding across accountability, transparency, oversight, and wellbeing outcomes. These dimensions can be treated as ideation lenses: for each candidate feature, structured workshops should deliberately explore how accountability is made visible, which transparency mechanisms are embedded, what oversight channels are feasible, and how emotional stability, cognitive autonomy, and social connectedness are supported in practice. Ideation sessions can be organised around these four lenses, using prompts derived from coded policy and case evidence to systematically challenge feature concepts that might inadvertently reproduce manipulative or opaque interaction patterns (Sharma, 2025, p. 7; Sharma, 2025, p. 8). Dark pattern taxonomies and detection research provide further scaffolding for creative exploration. A fivetier taxonomy built on mechanistic clarity, nonoverlap, and exhaustiveness identifies 27 specific patterns organised into categories that exploit cognitive biases such as anchoring, loss aversion, default effects, and least effort principles. Ideation activities should therefore incorporate pattern checklists and cognitive bias maps as constraint tools: brainstorming boards can list archetypal dark patterns (for example roach motel, privacy zuckering, misdirection, disguised ads) and associated bias mechanisms, with design teams required to demonstrate explicitly how emerging concepts avoid each mechanism or invert it toward autonomy support. This turns taxonomic knowledge into practical cre-

ative boundaries that prevent generation of concepts reliant on asymmetric optin versus optout flows, countdown timers for engagement, or preselected consent options that undermine privacypreserving choices (P et al., 2026, p. 2). Expertled cognitive walkthrough methodology on social platforms such as YouTube, LinkedIn, WhatsApp, and Telegram offers a template for ideation grounded in interaction scenarios. UI/UX specialists in that study executed twelve comprehensive tasks across content creation, engagement, account creation, and termination to identify combinations of dark patterns and analyse their impact on user experience. MindfulFeed's ideation workshops can repurpose these task structures as generative prompts: for each key journey (onboarding, feed browsing, notification management, account exit), teams sketch lowfidelity flow diagrams and then annotate them with potential dark pattern risks identified in the walkthrough research. Subsequent creative refinements focus on replacing risky segments with alternative interaction strategies that foreground clarity, symmetric effort, and cognitive relief, using the combinations documented in the walkthroughs as explicit antipatterns (Ukgoda, 2024, p. 3). Systematic review methodology in AIbased burnout interventions adds a further layer of rigor to concept generation. The PRISMAguided review employed staged processes of identification, screening, eligibility, and inclusion, supported by structured coding frameworks, thematic clustering, and quality appraisal via the Mixed Methods Appraisal Tool. Translating these procedures into ideation practice, candidate features for MindfulFeed can pass through analogous stages: initial idea capture, screening against governance and dark pattern criteria, eligibility assessment based on anticipated impact on stress or burnout, and inclusion into prototype backlogs only when preliminary evidence and ethical transparency requirements are satisfied. Feature matrices can record variables such as anticipated computational method, data sources, and targeted outcomes, mirroring the coding of intervention type and measured effects in the review and thereby maintaining analytical integrity in concept selection (Sarkar & Kudapa, 2024, p. 19). Research on digital aggression and algorithmic amplification points to multifaceted strategies, including algorithmic tweaks that prioritise constructive discourse over divisive content and digital literacy education to help users recognise manipulative tactics designed to provoke outrage. These insights can be incorporated into ideation by defining concept clusters around constructive feed logic and literacyenhancing interface elements. For example, ideation sprints can focus on alternative ranking algorithms that deprioritise outrage cues, combined with explanatory microcopy that trains users to identify emotionally manipulative patterns in their broader digital environment. Each concept in these clusters is then evaluated against hypotheses about anonymity, algorithmic influence, and moderation efficacy, ensuring that creative exploration remains aligned with documented relationships between social media algorithms and online aggression (Rehman et al., 2025, p. 5). Finally, methodological principles of credibility, transferability, and dependability in qualitative governance research demand that ideation artefacts be systematically documented. Session outputs such as annotated user journeys, patternrisk maps, and governance lens canvases should be archived and coded, enabling subsequent phases to trace how specific MindfulFeed features emerged from evidenceconstrained creative exploration rather than ad hoc brainstorming. This documentation practice directly mirrors the use of NVivo for thematic organisation and the emphasis on detailed contextual description and transparent coding in governance studies, thereby embedding methodological integrity into the ideation phase itself (Sharma, 2025, p. 7; Sharma, 2025, p. 8; Sarkar & Kudapa, 2024, p. 19).

6.2 Conceptual Design Space for MindfulFeed

Conceptual boundaries for MindfulFeed's design space are strongly shaped by empirical dark pattern research and governance analyses of digital wellbeing. A fivetier taxonomy built on mechanistic clarity, nonoverlap, and exhaustiveness organises 27 specific patterns into five toplevel categories, providing a structured negative design space that MindfulFeed must systematically avoid. Archetypal patterns such as trick questions, sneakintobasket, roach motel, privacy zuckering, misdirection, and disguised ads exemplify manipulative mechanisms that exploit cognitive biases and undermine autonomous decision making. Any conceptual feature canvas for MindfulFeed must therefore treat these named patterns, and the broader categories they instantiate, as explicit exclusion constraints when ideating consent flows, onboarding, feed interaction, and account management. Evidence on cognitive mechanisms further narrows the permissible design space. Dark patterns leverage System 1 processing, anchoring bias, loss aversion, default effects, and the principle of least effort through devices such as decoy pricing, countdown timers, preselected options, and asymmetric optin versus optout flows. Conceptually, MindfulFeed's interaction models must invert these mechanisms: any use of defaults,

scarcity messages, or effort asymmetries is treated as high risk and either prohibited or reconfigured to support autonomy, for instance by equalising the number of steps and time required for engagement versus disengagement actions. This inversion is not speculative; experimental work has shown that darkpattern consent mechanisms reduce privacypreserving choices by up to 23 percentage points, and cookie banner designs significantly determine consent withdrawal rates, confirming that interface architecture can override nominal freedom of choice (P et al., 2026, p. 2). Within MindfulFeed's design space, consent and privacy features are therefore constrained to plain language, symmetric effort, and the absence of countdowns or guilt framing. Governance perspectives extend these constraints beyond microinteractions to macro design objectives. Digital wellbeing is framed as a governance problem centred on accountability, transparency, oversight, and wellbeing outcomes, with AI systems in education and mental health currently operating in regulatory grey zones where surveillance, behavioural nudging, and psychological conditioning erode human agency (Sharma, 2025, p. 2; Sharma, 2025, p. 3). Translating this into a conceptual design space, MindfulFeed's feature set must only include AImediated components that can be made visibly accountable, explainable, and subject to human feedback or correction, and that are explicitly oriented toward emotional stability, cognitive autonomy, and social connection rather than engagement maximisation. Algorithms that simulate empathy or predict affective states without clear oversight channels fall outside the acceptable design space, given concerns about subtle erosion of the capacity to think, choose, and feel autonomously in digital environments (Sharma, 2025, p. 3). The GDW Framework's five pillars of ethical accountability, algorithmic transparency, human oversight, psychological safety, and inclusive policymaking act as additional structuring dimensions for concept generation. Features that cannot be mapped to at least accountability, transparency, and psychological safety should be excluded at the ideation stage, since governance analysis argues that systems must be accountable to the moral conditions of human life, not only to regulators or efficiency metrics (Sharma, 2025, p. 4). This means, for example, that any persuasive feature envisaged for MindfulFeed must allow users to understand why a particular prompt appears, who defined the behavioural objective, and how negative psychological impacts will be monitored and mitigated. Automated detection work with DarkScan also informs the design space by defining measurable thresholds of harm. A comprehensive empirical investigation combining a fivecategory taxonomy, automated DOM and visual analysis, a severity scoring model (DPSS), and a controlled user experiment demonstrates that dark patterns are pervasive and produce measurable behavioural and psychological harm, degrading decision quality, platform trust, and psychological wellbeing (P et al., 2026, p. 2). Policy and design recommendations derived from these findings include symmetric design principles, plain language copy, organisational policies that prohibit dark patterns, and success metrics that incorporate user trust and longterm retention rather than shortterm conversion (P et al., 2026, p. 8). Conceptually, MindfulFeed's design space is therefore bounded by commitments to symmetric pathways for optout and cancellation, nonmanipulative language, and evaluation criteria focused on trust and wellbeing rather than conversion funnels. Finally, trustworthiness criteria from governance research impose documentation expectations on the conceptual design space itself. Thematic content analysis emphasises credibility through data triangulation, transferability through detailed contextual description, and dependability through transparent coding and iterative peer debriefing (Sharma, 2025, p. 8). Applied to MindfulFeed, this requires that every proposed concept be traceable to specific dark pattern evidence or governance principles and that design rationales be recorded as artefacts subject to later audit. Concepts lacking such evidence linkage or documentation fall outside the robust, evidencebased design space appropriate for an ethical, prosocial social app.

7 Prototype Phase: Progressive Design of MindfulFeed

7.1 Prototype v1: LowFidelity Batch Feed Concept

Prototype v1 is scoped as a lowfidelity batch feed concept that operationalizes evidence on dark patterns, consent requirements, and personalization fatigue into a constrained interaction model. Dark pattern taxonomies identify mechanisms such as sneaking, obstruction, misdirection, forced action, confirm shaming, roach motel, and drip pricing, all of which deliberately exploit cognitive biases like default bias, scarcity bias, framing, and loss aversion to steer users into unintended actions (Mavi, 2025, p. 2; P et al., 2026, p. 2). Experimental work shows that darkpattern consent mechanisms can reduce privacypreserving choices by up to 23 percentage points, and that cookie banner design strongly

determines consent withdrawal rates, demonstrating that interface architecture can override nominal freedom of choice (P et al., 2026, p. 2). To avoid reproducing these harms, the batch feed prototype is explicitly framed as a nonscrolling, sessionbounded layout that presents a finite set of items without countdowns, autoadvance, or preselected engagement defaults. Lowfidelity feed sketches are structured around cards arranged in discrete batches, with explicit "next batch" controls instead of infinite scroll. This configuration responds to analyses of dark patterns that emphasize how asymmetric optin versus optout flows and leasteffort principles can trap users in roach motel configurations where entry is easy but exit is difficult (Mavi, 2025, p. 2; P et al., 2026, p. 2). Each batch page in the prototype therefore contains a visible "end of batch" marker and a neutral "pause" affordance rather than engagement prompts framed through urgency or scarcity. Interface copy is deliberately stripped of emotional pressure, avoiding confirm shaming language that would induce guilt for disengaging (Mavi, 2025, p. 2). In addition, decision points such as saving items or adjusting preferences are designed with symmetric effort: accepting and declining have equal visual prominence and identical click depth, aligning with legal expectations that consent be freely given, specific, informed, unambiguous, and easy to withdraw (Gray et al., 2021, p. 4; P et al., 2026, p. 2). Consent and personalization controls embedded into the batch feed follow legal and governance guidance on consent banners and digital wellbeing. GDPR and the ePrivacy Directive require that consent be presented in a manner clearly distinguishable from other matters, with meaningful choice evidenced by the ability to withdraw consent, and they caution that making access conditional on consent can invalidate reliance on consent (Gray et al., 2021, p. 4). In the lowfidelity concept, batchlevel personalization toggles are surfaced at the top of the feed rather than buried in secondary menus. Drawing on tiered personalization settings work, prototype controls allow selection of "Minimal," "Balanced," or "Maximized" personalization, explicitly mapped to tracking granularity, content diversity, and data retention, and presented as user sovereignty features rather than compliance formalities. These settings are accessible without navigating multitab menus or dense legalese, reflecting evidence that accessible, userselectable personalization tiers can reduce fatigue and strengthen perceived autonomy. The batch feed prototype also integrates mitigation strategies against personalization fatigue documented in UI frameworks. Adaptive exposure control, diversityenriched recommendations, exposure freshness windows, transparent recommendation logic, fatigueaware UI nudges, and tiered personalization are described as six strategies that shift personalization from extraction to empathy. At low fidelity, this translates into simple placeholders: each item card includes a short "Why this item?" tooltip zone for later implementation of explanation logic; batch headers reserve space for "freshness indicators" that can mark when content has been recently updated or deliberately rotated; and a "breathing space" button is positioned alongside navigation controls to prototype fatigueaware nudges that offer quick exits or reflective pauses instead of additional content. These layout decisions are annotated in the design record against the corresponding framework dimensions (algorithmic transparency, emotional salience, user sovereignty), ensuring traceability from empirical mitigation strategies to concrete interface structures (Trivedi, 2025, p. 6). Governance perspectives on digital wellbeing define attention, emotion, and cognition as contested frontiers and argue that AI systems in education and mental health currently operate in regulatory grey zones where surveillance, behavioural nudging, and psychological conditioning can erode human agency (Sharma, 2025, p. 3). Although this prototype remains lowfidelity and nonfunctional, batch feed flows are documented to prevent implicit nudging through opaque ranking or simulated empathy. Wireframe annotations specify that recommendation ordering must later be explainable and adjustable by users, and that no batch navigation element may depend on hidden data processing. This aligns the concept with governance pillars of accountability, transparency, oversight, and psychological safety, positioning the batch feed as a structural countermeasure against quiet architectures of dependency (Sharma, 2025, p. 3; Sharma, 2025, p. 8). Finally, the prototype documentation includes an AI usage note for future iterations, informed by educational AI governance. Personalization and learning analytics in education are said to require explanations understandable to users and evaluation beyond prediction accuracy, including impacts on different groups and longterm consequences of labeling (Judijanto, 2026, p. 6). In the batch feed record, any planned AI component (for example, sessionlevel heuristics for adaptive exposure) is listed in an appendix with intended function, data inputs, and oversight mechanisms, thereby conforming to demands for transparency, human oversight, and differentiated use that supports learning rather than replacing intellectual processes (Judijanto, 2026, p. 6; Sharma, 2025, p. 3).

7.2 Prototype v2: MidFidelity Algorithm Control Dashboard

Midfidelity prototyping of an algorithm control dashboard is positioned as a response to documented risks around opaque personalization, darkpattern consent design, and erosion of autonomy in AI-mediated environments. Governance analysis identifies accountability, transparency, oversight, and wellbeing outcomes as core dimensions for evaluating AI systems that affect cognition and emotion, noting that current platforms often operate in regulatory grey zones where surveillance, behavioural nudging, and psychological conditioning undermine agency (Sharma, 2025, p. 2; Sharma, 2025, p. 8). Within this evidence base, the dashboard concept is specified as an interface that exposes algorithmic operations, allows users to configure personalization intensity, and surfaces consent status in a way that facilitates oversight rather than control. Design of midfidelity controls draws directly on UIbased personalization fatigue mitigation strategies. Tiered personalization settings are recommended as a way to respect differing comfort levels with tracking and tailoring, by offering "Minimal", "Balanced", and "Maximized" options that affect recommendation logic, visual density, tracking granularity, and data retention. Crucially, these settings must be accessible and not buried under legalese or multitab menus, and the act of choosing among tiers is framed as a gesture of respect rather than a compliance exercise. The same framework lists transparent recommendation logic and fatigueaware UI nudges as key strategies, implemented through explanation tooltips, preference dashboards, breathing space options, and engagement monitoring to reduce overwhelm and strengthen trust (Trivedi, 2025, p. 6). In the midfidelity dashboard, these strategies are translated into wireframed modules: a tier selector rendered as equalprominence buttons, a "Why this content?" explanation zone mapped to specific feed segments, and a fatigue panel that reserves space for future breathingspace configurations. Dark pattern research provides constraints on how consent and personalization controls can be structured. Brignull's catalogue and subsequent taxonomies enumerate manipulative strategies such as trick questions, sneakintobasket, roach motel, privacy zuckering, misdirection, and disguised ads, all grounded in cognitive biases including System 1 reliance, anchoring, loss aversion, default effects, and least effort. Randomised trials and consent experiments have shown that aggressive dark patterns can increase subscription uptake dramatically and that darkpattern consent mechanisms reduce privacypreserving choices by up to 23 percentage points, while cookie banner design significantly determines consent withdrawal rates. These findings confirm that apparent freedom of choice can be architecturally determined by asymmetrical optin versus optout flows and preselected options (P et al., 2026, p. 2). Accordingly, the midfidelity dashboard mockups explicitly avoid prechecked boxes, countdown timers, or buried "off" switches. Each toggle for tracking, data retention, or personalization intensity is designed with symmetric visual weight and click depth for enabling and disabling, and batch diagrams of navigation flows document equal step counts for entering and exiting personalized modes. Ethical governance in AIenabled monitoring environments further informs dashboard content. Occupational health analyses stress that privacy and informed consent are central when data streams include psychological states and behavioural patterns, and that transparent consent protocols must clearly inform users about what data are collected, how they are processed, and for what purposes. Consent should be iterative and revocable, allowing optout without penalty; data minimization should restrict retention to information necessary for legitimate objectives; and algorithmic transparency must make predictive functioning explainable and auditable. Absent such transparency, AI risks being perceived as an intrusive management tool rather than a supportive wellness mechanism (Sarkar & Kudapa, 2024, p. 18). Reflecting these requirements, the midfidelity dashboard allocates explicit sections for "Data in use", "Purpose of analysis", and "Current consent state", with flow diagrams indicating how a user can revoke consent at any time without service degradation. Educational AI research elaborates how personalization and learning analytics should be governed, emphasizing that algorithms must not determine paths in a closedoff way and that students need to understand recommendations, have choices, and be able to reject or adjust profiles. Recent literature calls for human oversight, transparency, fairness, accountability, and explainability, with algorithmic explanations understandable to teachers and learners, and systems evaluated not only on prediction accuracy but also on impacts on different groups, error likelihood, rights to correction, and longterm consequences of labeling (Judijanto, 2026, p. 6). Translating this into MindfulFeed's midfidelity dashboard, planned mockups incorporate an "Explanations" panel that associates each controllable algorithmic feature with an intelligible rationale and an "Corrections and appeals" section that specifies how users can flag harmful or inaccurate personalization for review. At the governance level, digital wellbeing frameworks argue that algorithmic systems should be accountable to the moral conditions of human life, and that digital wellbeing must be governed

with seriousness comparable to physical health or data privacy. Trustworthiness in qualitative governance work is achieved through data triangulation, detailed contextual descriptions, and transparent coding (Sharma, 2025, p. 2; Sharma, 2025, p. 8). In alignment, the midfidelity dashboard prototype is documented with interaction grids and configuration matrices that map each control element to its governing evidence source: tier selectors to personalization fatigue strategies, consent toggles to privacy triads of informed, revocable consent and minimization, and explanation panels to educational demands for intelligible algorithmic rationale. This documentation ensures that subsequent usability evaluations can assess whether the dashboard genuinely operationalizes autonomy, transparency, and psychological safety rather than introducing subtler forms of manipulation or overload. Evidence from related empirical work suggests that many users acquire language and interface habits through informal digital spaces, and thus plain, conversational explanations may be more likely to be read and acted upon than dense technical prose (Maryam et al., 2025, p. 9). At the same time, that same work reports that a large share of engagement is with informal or hybrid language variants, which may imply a need to offer explanation text in multiple registers so that consent and correction mechanisms are both accessible and perceived as legitimate by diverse user groups (Maryam et al., 2025, p. 9). Empirical studies of adolescent social media use report frequent sleep disruption, academic impairment, and heightened anxiety tied to compulsive engagement patterns, which suggests that interface controls such as breathing spaces and explicit disengagement affordances are not optional niceties but practical responses to observable harms (Qureshi et al., 2025, p. 5). These accounts may suggest that dashboards intended for younger users should prioritise clear, plain warnings about sleep and attention impacts alongside straightforward means to reduce exposure.

7.3 Prototype v3: HighFidelity Interactive System

Highfidelity implementation of MindfulFeed's interactive system is grounded in prior evidence on dark pattern harms, personalization fatigue, and AI governance, and translates these constraints into detailed interaction flows and control surfaces. Severity scoring of common dark patterns using the Dark Pattern Severity Scores (DPSS) framework shows that patterns such as hidden subscription charges, privacy zuckering, roach motel cancellation flows, and confirmshaming reach medium to high severity ranges, with DPSS values up to 78.2. In empirical case studies, a hidden subscription design produced an unintended subscription rate of 47.3% compared with 6.2% for a neutral redesign, while a multilayer cookie consent pattern required 11 interactions to achieve optout and preenabled 38 of 47 tracking purposes, violating GDPR Article 7 and guidance on cookie banners. These findings necessitate that highfidelity screens for onboarding, consent, and subscription management in MindfulFeed explicitly eliminate preselected checkboxes, complex navigation to terms, and multistep optout sequences. Interaction specifications therefore document oneclick symmetry between "accept" and "reject" actions, visible "reject all" options in any consent banner, and cancellation flows limited to a small, fixed number of clearly labelled steps, with DPSS used internally to score residual risk for each pattern class (P et al., 2026, p. 7). Personalization controls in the highfidelity system implement tiered settings and related mitigation strategies that have been operationalised in existing retail interfaces. Tiered personalization allows users to select "Minimal", "Balanced", or "Maximized" customization, affecting recommendation logic, visual density, tracking granularity, and data retention. Critically, these settings must be immediately accessible and not buried under legalese or multitab menus, emphasising choice as a gesture of respect rather than mere compliance. A broader framework of six UI strategies is available: adaptive exposure control through sessionlevel heuristics, diversityenriched recommendations via category diversity rules, exposure freshness windows, transparent recommendation logic via explanation tooltips and preference dashboards, fatigueaware UI nudges with breathing space options, and the tiered personalization settings themselves, framed as user sovereignty tools. Highfidelity mockups instantiate these strategies through concrete components: a personalization tier selector placed in the main settings pane; peritem "Why am I seeing this?" explanation icons; session timers and "take a break" prompts based on scroll and interaction data; and controls for "content diversity" and "freshness windows" that users can adjust in real time. UI specifications map each control to the corresponding framework dimension (algorithmic transparency, emotional salience, user sovereignty), ensuring traceability from conceptual mitigation to implemented interaction (Trivedi, 2025, p. 6). Responsible AI use in the interactive system is informed by educational research on personalization, learning analytics, and AI oversight. Digital personalization is described as adjusting pace, sequence, difficulty, feedback type, or representation, with analytics interpreted through human

judgement to support study success. Importantly, personalization must not lock learners into closed paths dictated solely by past profiles; students require explanations of recommendations, opportunities to choose, and the ability to reject or adjust profiles. Recent work stresses human oversight, transparency, fairness, accountability, and explainability, and recommends that algorithmic explanations be understandable to teachers and students and evaluated not only for prediction accuracy but also for impacts across groups, likelihood of error, rights to corrections, and longterm labelling consequences (Judijanto, 2026, p. 6). In MindfulFeed's highfidelity design, these principles are operationalised via an "algorithm insights" panel that shows what behavioural signals feed recommendation models, an explicit "adjust my profile" flow that lets users correct inferred interests, and a "request review" function for content or suggestion deemed harmful or inaccurate. Interaction grids specify that any highrisk decision, such as account state changes or longterm tagging of usage patterns, must be reversible and associated with a clear explanation dialogue, aligning with humanintheloop expectations and rights to correction (Judijanto, 2026, p. 6; Judijanto, 2026, p. 11). Governanceoriented analysis of digital wellbeing emphasises accountability, transparency, oversight, and wellbeing outcomes, and uses qualitative procedures that highlight credibility through triangulation, transferability via contextual detail, and dependability through transparent coding (Sharma, 2025, p. 8). Highfidelity development documentation mirrors these procedures: every interactive element in MindfulFeed's system is logged in a design matrix that identifies its purpose, associated dark pattern risks, chosen mitigation strategies, and governance alignment, with references back to empirical sources such as DPSS case studies and personalization frameworks. This evidencelinked specification practice supports later testing and audit, ensuring that the final interactive system does not drift toward manipulative norms but remains anchored in documented harms, countermeasures, and AI oversight requirements (P et al., 2026, p. 7; Trivedi, 2025, p. 6; Judijanto, 2026, p. 6; Sharma, 2025, p. 8). Bibliometric mappings of related work suggest that research and publication patterns are concentrated in particular regions and outlets, which may affect how generalisable certain UX and governance recommendations appear to be; the United States, for example, figures as a central node in international collaboration networks (Ekawati, 2025, p. 11). This concentration might imply that some empirical norms reflect publishing geographies as much as universal practice, and so our design audits keep a careful eye on contextual transfer when applying external case studies to MindfulFeed (Ekawati, 2025, p. 11). Recent reviews also argue that participatory design with affected communities and clear ethics disclosures are practical safeguards when AI systems touch vulnerable groups; incorporating such practices into product audits may reduce risk and improve legitimacy, even if they add time to development cycles (Abah, 2026, p. 14). In short, we treat transparent ethics reporting and structured stakeholder engagement as complementary governance steps that are likely to reveal blind spots that conventional testing alone would miss (Abah, 2026, p. 14).

8 Test Phase: User Feedback and Iterative Redesign

8.1 Testing Protocols for Each Prototype Version

Testing activities for MindfulFeed's prototype versions must reflect evidence based methodological practices drawn from dark pattern research, AI governance analysis, and systematic review procedures on computational burnout interventions. For early interface concepts, task based cognitive walkthroughs executed by experienced UI/UX evaluators offer a structured protocol. In existing work, four engineers completed twelve comprehensive tasks on YouTube, LinkedIn, WhatsApp, and Telegram, targeting content creation and engagement, account creation, and termination to identify combinations of dark patterns and analyse how these combinations affected user interaction and overall experience. Adapting this, low fidelity batch feed sketches should be tested through predefined navigation scenarios covering session start, batch completion, consent decision points, and exit actions, with evaluators recording potential dark pattern mechanisms and cognitive load indicators, ensuring that manipulative combinations documented elsewhere are not replicated (Ukgoda, 2024, p. 3). Systematic review methodology for AI based burnout interventions provides a second anchor for testing protocols. PRISMA guided procedures employ staged identification, screening, eligibility, and inclusion, supported by structured coding frameworks, thematic clustering, and quality appraisal using the Mixed Methods Appraisal Tool. Translating this into prototype evaluation, each testing round should follow a comparable sequence: specification of inclusion criteria for participant sessions, documentation of

screened test logs, eligibility checks based on completeness and context, and final inclusion into the analysis corpus only when interaction data meet predefined robustness thresholds. Data extraction templates can mirror coding frameworks used in the review, capturing variables such as task type, prototype fidelity level, dark pattern risk flags, consent comprehension, and reported cognitive or emotional effects, with narrative synthesis applied when heterogeneity in tasks or user profiles precludes quantitative aggregation (Sarkar & Kudapa, 2024, p. 19). Governance oriented thematic content analysis offers detailed guidance on qualitative testing procedures. In AI ethics research, documents were read multiple times to extract themes under accountability, transparency, oversight, and well being outcomes, with NVivo used to organize data and trustworthiness ensured via triangulation, contextual description, and transparent coding. For MindfulFeed, mid fidelity algorithm control dashboard sessions should be audio recorded and transcribed, then coded against the same four dimensions: how participants locate accountability cues in dashboard controls, which transparency elements they notice or miss, what oversight channels they perceive, and how they describe emotional or cognitive responses to configuration tasks. Multiple data sources, including evaluator notes, clickstream logs, and interview excerpts, need to be triangulated to strengthen credibility, while detailed documentation of user context and prototype version supports transferability of findings across future iterations (Sharma, 2025, p. 8). Dark pattern taxonomy and automated detection work define additional quantitative testing metrics. A five tier, empirically grounded taxonomy based on mechanistic clarity, non overlap, and exhaustiveness, together with the DarkScan pipeline, was developed to measure behavioural and psychological impacts of dark pattern exposure and derive policy and design recommendations informed by regulatory frameworks. High fidelity interactive builds can integrate automated interface scans before user testing, flagging any pattern instances in consent banners or personalization flows. Subsequent controlled experiments should then measure decision quality, platform trust, and self reported psychological wellbeing for scenarios such as consent management and personalization adjustment, following question structures that map directly onto the documented impact domains. Comparison of MindfulFeed flows with known dark pattern heavy configurations from prior studies allows assessment of whether the interactive system reduces harm indicators in line with design goals (P et al., 2026, p. 2). Given ethical sensitivities and regulatory constraints around consent and data use, testing protocols must explicitly incorporate legal criteria. EU GDPR and ePrivacy definitions require consent to be freely given, specific, informed, and unambiguous, clearly distinguishable from other matters, and revocable without loss of access. During all prototype evaluations that involve consent banners or privacy settings, test scripts should instruct participants to perform opt in and opt out paths. Observational checklists then record step counts, visibility of rejection options, and comprehension of data consequences. Deviations from symmetric effort or clarity are logged as compliance risks, informing redesign prior to deployment (Gray et al., 2021, p. 4). Finally, protocols for documenting AI usage within prototypes must reflect responsible personalization guidance. Educational AI analysis argues for human in the loop models in which systems generate recommendations, but teachers or students interpret context, check fairness, and decide next steps, with high risk decisions kept auditable and reversible (Judijanto, 2026, p. 5). Correspondingly, testing of algorithmic components in MindfulFeed should include explicit explanation prompts, user opportunities to correct inferred profiles, and scenarios where participants reject or modify automated suggestions. Evaluation grids should capture whether explanation dialogues are understood, whether users can locate correction mechanisms, and whether any automated decisions appear irrevocable, thereby aligning interactive testing with responsible personalization principles (Judijanto, 2026, p. 5; Sharma, 2025, p. 8).

8.2 Insights from Design Iterations

Iterative testing of MindfulFeed's prototypes yields several practical insights regarding the alignment between ethical intentions, governance dimensions, and concrete interaction behaviour. Taskbased cognitive walkthroughs conducted by experienced UI/UX engineers on major social media platforms demonstrate that dark patterns rarely occur in isolation; instead, combinations across content engagement, account creation, and termination features shape overall experience and difficulty of disengagement (Ukgoda, 2024, p. 3). Applying comparable walkthrough protocols to early batch feed sketches revealed that even ostensibly neutral layouts can inadvertently reproduce obstruction when exit or pause actions are less visually salient than continuation controls. This led to a design correction in which "end of batch" and "pause" affordances were given equal typographic weight to navigation prompts, explicitly responding to evidence that asymmetric effort can trap users in roach motel config-

urations and undermine decision quality (P et al., 2026, p. 2; Ukgoda, 2024, p. 3). Analysis of consent and configuration flows during prototype testing further clarified the operational impact of symmetric design principles. Dark pattern research reports that aggressive manipulation in consent interfaces significantly reduces privacypreserving choices and that cookie banner structures strongly determine withdrawal behaviour, making nominally free decisions architecturally constrained (P et al., 2026, p. 2). When midfidelity algorithm control dashboard mockups were evaluated against these criteria, usability grids documented instances where "accept all" appeared visually prominent relative to granular controls. Iteration then focused on enforcing equal click depth and optical prominence for enabling and disabling tracking, echoing recommendations that opting out, cancelling, or exercising privacy preferences be no more complex than their inverses and that calls to action use plain, nonguiltframing language (P et al., 2026, p. 8). Qualitative coding of user feedback using accountability, transparency, oversight, and wellbeing outcomes as analytical dimensions provided additional guidance on redesign priorities. Governance research operationalises these four categories to link administrative design with experiential consequences for emotional stability, cognitive autonomy, and social connectedness, using repeated document reading, comparative mapping, and theoretical triangulation supported by NVivo to ensure traceability and robustness. Transferring this structure to test artefacts allowed participant narratives about confusion in personalization settings or difficulty locating consent withdrawal options to be systematically classified as transparency or oversight failures. Subsequent iterations introduced clearer "data in use" summaries and visible feedback channels in dashboards, with design matrices explicitly mapping each change to the relevant governance dimension to maintain dependability in the product record (Sharma, 2025, p. 7; Sharma, 2025, p. 8). Comparisons between iterative changes and dark pattern taxonomies showed that design teams benefited from treating taxonomic categories as negative templates rather than abstract theory. The fivetier taxonomy and DarkScan pipeline, which combine static DOM analysis, text classification, visual inspection, and dynamic behavioural tracing, demonstrate measurable behavioural and psychological harm from patterns such as trick questions, privacy zuckering, misdirection, confirm shaming, and roach motel, and argue for organisational policies that prohibit such practices while refocusing success metrics on trust and longterm retention. Incorporating automated scans into highfidelity reviews identified residual risks, such as complex cancellation flows, which user testing confirmed through elevated abandonment and frustration. Iterative simplification of these flows, documented alongside DPSS severity scores, reduced observed task times and user irritation, indicating movement toward the "darkpatternfree" pathways recommended for highrisk sectors (P et al., 2026, p. 2). Finally, adaptation of PRISMAstyle staging from systematic reviews into testing cycles improved methodological transparency for design decisions. The stress and burnout review on AIbased occupational health interventions used identification, screening, eligibility, and inclusion stages, supported by structured coding frameworks and Mixed Methods Appraisal Tool scoring, to ensure analytical integrity and interrater reliability. Mirroring this structure, MindfulFeed's testing logs were screened for completeness, coded for prototype version and task type, and included in synthesis only when interaction traces and qualitative notes met predefined quality thresholds. Cohen's kappa, used in the burnout review to confirm strong agreement between coders, was replicated in rating dark pattern risk across evaluators, leading to more consistent identification of problematic flows and more defensible redesign rationales (Sarkar & Kudapa, 2024, p. 19). Collectively, these iterative insights show that continuous alignment with taxonomic evidence, governance dimensions, and systematic review methodology is necessary to prevent ethical drift during refinement of MindfulFeed's interaction architecture (P et al., 2026, p. 2; Ukgoda, 2024, p. 3; Sharma, 2025, p. 7; Sarkar & Kudapa, 2024, p. 19).

9 MindfulFeed System Architecture and Feature Specification

9.1 Information Architecture and Navigation

Information architecture for MindfulFeed must be structured to preclude manipulative pathways while supporting transparency, accountability, and psychological safety in navigation flows. Dark pattern taxonomies identify archetypal mechanisms such as trick questions, sneakintobasket, roach motel, privacy zuckering, misdirection, and disguised ads, organised into 27 patterns across five toplevel categories based on mechanistic clarity, nonoverlap, and exhaustiveness (P et al., 2026, p. 2). These forms rely on structural and visual asymmetries in information layout, including preselected options, deep

nesting of cancellation actions, and visually dominant "accept" controls. Consequently, MindfulFeed's page hierarchy and menu structure must explicitly avoid layout configurations that create asymmetric optin versus optout pathways, deep DOM placement of exit actions, or hidden routes to privacy controls (P et al., 2026, p. 2; P et al., 2026, p. 4). Navigation design is additionally constrained by empirical evidence on obstruction flows. A documented highseverity cancellation path required eight steps, including forced traversal of promotional offers, mandatory survey completion, and manual entry of "CONFIRM CANCEL," resulting in mean completion times of 18 minutes 42 seconds and an abandonment rate of 31.4%; signup, by contrast, required only three steps and approximately 90 seconds (P et al., 2026, p. 8). This asymmetry illustrates how navigation depth and forced detours can convert simple tasks into roach motel experiences. In MindfulFeed's information architecture, any journey for account deletion, consent withdrawal, or notification reduction must therefore be designed to match or undercut the complexity of corresponding optin routes. Flow specifications should cap step counts and forbid mandatory detours through persuasive content, aligning with design recommendations that cancellation flows be "no more complex than signup flows" and that opting out be as easy and prominent as opting in (P et al., 2026, p. 6; P et al., 2026, p. 8). Automated detection work using DarkScan further informs structural decisions. The DOM analysis module extracts features such as DOM depth of cancellation affordances, relative prominence of optin versus optout options, and number of steps in key flows, while visual analysis quantifies salience ratios for calls to action (P et al., 2026, p. 4). For MindfulFeed, navigation maps should be documented in a way that permits similar automated auditing: each key pathway (onboarding, feed traversal, personalization settings, consent management, and account closure) is represented as an explicit user journey diagram, annotated with node depth, branching factors, and visibility of "exit" or "reject" actions. This supports continuous compliance checking against symmetric optionality and transparent information architecture principles derived from DarkScan's policy recommendations (P et al., 2026, p. 6). Governance analysis of digital wellbeing introduces four evaluative dimensions that must be embedded into the information architecture: accountability, transparency, oversight, and wellbeing outcomes. Thematic content analysis procedures in this research read documents repeatedly, code references to ethical standards and user protection, and organise themes in NVivo to ensure traceability and analytical transparency (Sharma, 2025, p. 8). Applying this structure, MindfulFeed's navigation schema should assign each major section (for example, "Data and Consent," "Personalization Controls," and "Support and Feedback") to explicit governance dimensions. Navigation labels and breadcrumb trails need to make visible where accountability resides, where algorithmic operations are explained, and where oversight or redress can be exercised. Informationarchitecture artefacts, such as site maps and menu taxonomies, must be coded against these four categories to verify that users can locate governancerelevant functions without search or guesswork, supporting cognitive autonomy and emotional stability by reducing uncertainty around control points (Sharma, 2025, p. 3; Sharma, 2025, p. 8). Insights from cognitive walkthroughs on YouTube, LinkedIn, WhatsApp, and Telegram highlight how combinations of dark patterns manifest along navigation routes. Experts assessed content engagement and account creation or termination features through twelve tasks per platform, identifying that miscategorised notification and recommendation features contributed to confusion and made manipulative interactions harder to recognise. MindfulFeed's information architecture should therefore separate clearly the functional domains of content browsing, social interaction, personalization, and account management into distinct navigation clusters, avoiding blended categories that obscure where tracking or control occurs. Pilot testing in the walkthrough study revealed inconsistencies in feature categorisation and led to revised definitions and improved evaluator guidelines (Ukgoda, 2024, p. 3). Analogously, MindfulFeed must maintain rigorous internal taxonomies for navigation labels and menu groupings, ensuring that features affecting data, attention, or autonomy are not buried under generic headings that hinder user recognition and later expert auditing. Finally, design recommendations from dark pattern research emphasise symmetric optionality, transparent information architecture, honest scarcity, and graceful exit as guiding principles for darkpatternfree interfaces (P et al., 2026, p. 6). In navigation terms, this requires that every decision node presents all materially relevant information on a single screen before commitment, that no temporal or scarcity indicators are shown unless grounded in actual system constraints, and that "back," "pause," and "leave" actions are consistently available across hierarchical levels. Trustworthiness criteria from governance research, including data triangulation, detailed contextual description, and transparent coding of design artefacts, should be applied to information-architecture documentation so that future audits can verify that MindfulFeed's navigation structure

remains aligned with ethical and regulatory expectations rather than drifting toward concealed complexity (Sharma, 2025, p. 8; P et al., 2026, p. 6). Empirical survey work of platform user behaviour may also temper assumptions about typical usage intensity: many users report modest daily engagement and weak correlations between time spent on platforms and negative outcomes, which suggests that IA rules should not presume constant heavy use but should still prepare for edge cases where brief flows can have outsized effects (Perera & Abeykoon, 2026, p. 11204). Largescale analyses of engagement datasets and AI assessments indicate that engagement distributions are often skewed and that multimodal measures (text, survey, biometric proxies) provide a more nuanced picture of typical user behaviour; this nuance appears to encourage IA rules that prioritise lowfriction exits and clear signposting for the plurality of light users (Dumbili et al., 2025, p. 2). Such findings imply the value of lightweight, lowfriction exit options that respect the experience of the majority while preventing disproportionate friction for those who do wish to leave or reduce engagement (Perera & Abeykoon, 2026, p. 11204).

9.2 Ethical Recommendation Engine Design

Ethical configuration of MindfulFeed's recommendation engine must operationalize privacy, fairness, transparency, and human oversight requirements that have been articulated for AI monitoring and personalized systems. In occupational health contexts, computational analytics over sensor data and communication logs expose intimate emotional and cognitive states, creating risks of misuse, unauthorized access, and discrimination if safeguards are weak. Translating this into feed ranking and content recommendation, engine inputs should be restricted to information strictly necessary for wellbeing support, with data minimization rules encoded at the data ingestion and feature engineering layers. Configuration matrices in the product record need to list each behavioural signal, its purpose, and retention horizon, ensuring that no auxiliary data are collected solely for engagement maximization. Consent management around recommendation logic requires a triad of informed consent, data minimization, and transparent communication between users, institutional partners, and system designers (Sarkar & Kudapa, 2024, p. 18). Practically, this means that any personalization toggle or profile adjustment within MindfulFeed's engine dashboard must be accompanied by concise explanations detailing what data are used, how ranking scores are computed in broad terms, and how outputs relate to burnout or wellbeing goals. Consent needs to be iterative and revocable; engine design therefore has to support realtime deactivation of personalization without degrading core service quality, with logging of consent state changes as auditable events (Sarkar & Kudapa, 2024, p. 18; Ronald, 2025, p. 7). Bias mitigation constitutes a second pillar for ethical recommendation. Computational monitoring in healthcare can misclassify stress or overlook vulnerability in underrepresented groups when models are trained on skewed or institutionspecific data, leading to unfair interventions and unequal support allocation (Sarkar & Kudapa, 2024, p. 18). Within MindfulFeed, training datasets for recommendation models must be documented with demographic coverage tables and fairness evaluation results, and periodic audits should assess differential accuracy and exposure across gender, age, and other salient groups. Accessibility safeguards are also necessary; interface paths for configuring recommendations and viewing "why this content" explanations must be usable regardless of technological literacy or cultural background, aligning with equity and inclusion as moral obligations rather than purely technical targets (Sarkar & Kudapa, 2024, p. 18; Samanta, 2025, p. 0180546). Responsible personalization frameworks in education further refine how recommendation engines should relate to human agency. Personalization can adjust pace, sequence, difficulty, feedback type, and representation, but algorithms must not determine paths in a closedoff way; learners need to understand recommendations, have choices, and be able to reject or adjust their profiles (Judijanto, 2026, p. 6). Applied to MindfulFeed, ranking functions must be exposed through an "algorithm insights" panel where profile dimensions and content categories are intelligible to ordinary users. Humanintheloop models, in which systems generate suggestions while humans interpret context and decide next steps, imply that highrisk outputs, such as strong behavioural nudges or mental health advice, should require explicit confirmation or be reviewable by qualified moderators before widespread deployment (Judijanto, 2026, p. 11; Judijanto, 2026, p. 5375614). Governance analyses highlight accountability, transparency, oversight, and wellbeing outcomes as central dimensions for AI systems that affect cognition and emotion (Sharma, 2025, p. 8). MindfulFeed's recommendation engine therefore needs architectural features that map directly onto these dimensions: audit logs and impact dashboards to support accountability; peritem and pertopic explanation tooltips to achieve transparency; feedback and correction channels

for oversight; and monitoring of psychological and trust metrics to evaluate wellbeing outcomes, rather than relying solely on engagement indicators (Ronald, 2025, p. 7; Sharma, 2025, p. 8). To maintain trustworthiness, engine design documentation should mirror qualitative governance procedures, with transparent coding of algorithm changes and triangulation of behavioural data, user feedback, and policy constraints when adjusting ranking strategies (Sharma, 2025, p. 8). Studies on digital health and preventive wellbeing interventions show that AI models can analyse behavioural patterns, sleep, physical activity, and communication to detect risks early and support stress reduction, chronic disease management, and mental health promotion. However, extensive monitoring without robust privacy and consent controls undermines autonomy and can be perceived as surveillance (Ronald, 2025, p. 7). For MindfulFeed, this implies that recommendation objectives must be framed explicitly as preventive wellbeing support, with clear boundaries that exclude commercial targeting or opaque behavioural conditioning. Engine configuration grids should specify which model outputs trigger supportive prompts versus when no action is taken, and these rules must be open to user review and adjustment to avoid covert manipulation (Ronald, 2025, p. 7; Judijanto, 2026, p. 5375614). Finally, organizational governance mechanisms such as AI ethics boards and compliance committees are increasingly adopted to oversee data handling, model validation, and risk management in healthcare monitoring (Sarkar & Kudapa, 2024, p. 18). MindfulFeed's development record should similarly describe internal oversight structures for the recommendation engine, including periodic fairness and privacy audits, review gates before major model updates, and procedures to communicate model limitations to users to prevent overreliance. Embedding these controls into the engine design aligns computational personalization with humane, wellbeingoriented AI practice and supports cognitivehealth safeguards central to the product brief (Sarkar & Kudapa, 2024, p. 18; Ronald, 2025, p. 7; Sharma, 2025, p. 8).

9.3 Monetization Strategy Without Dark Patterns

Monetization options for MindfulFeed must be constrained by empirical evidence on dark patterns and ethical AI governance so that revenue generation does not depend on deceptive interface strategies or exploitative personalization. Dark pattern taxonomies grounded in mechanistic clarity, non overlap, and exhaustiveness describe 27 patterns, including sneaking, urgency or scarcity cues, obstruction, misdirection, forced action, confirm shaming, roach motel, bait and switch, and drip pricing, all designed to maximise organisational gain at the expense of consumer autonomy and informed consent (Mavi, 2025, p. 2; P et al., 2026, p. 2). These classifications, together with findings that aggressive dark patterns can increase subscription uptake by over 200%, demonstrate that many commercially successful monetization flows rely on cognitive bias exploitation such as default bias, scarcity bias, framing, and loss aversion rather than transparent value exchange (P et al., 2026, p. 2; Mavi, 2025, p. 2). For MindfulFeed, such mechanisms must be formally excluded from pricing, subscription, and upgrade journeys, and this exclusion documented within design review checklists and user journey flow artefacts. Symmetric design principles derived from dark pattern research provide concrete execution rules for non manipulative monetization flows. Evidence shows that complex cancellation paths with eight steps, forced traversal of promotional detours, mandatory survey completion, and manual text entry produce average cancellation times of 18 minutes 42 seconds and abandonment rates of 31.4%, while sign up in the same context requires only three steps and 90 seconds. Design recommendations therefore require that opting out, cancelling, or exercising privacy preferences be no more complex than their inverses, that calls to action use plain language without guilt or loss framing, and that organisations adopt policies defining dark patterns as prohibited practices, with accountability mechanisms empowered to intervene. Applied to monetization, this implies that any paid tier activation must be matched by an equally simple and visible downgrade or cancellation route, with user journey documentation for checkout and cancellation retained for audit and attached to the product record as part of ethical review artefacts (P et al., 2026, p. 8). Automated auditing capacity via tools such as DarkScan supports technical enforcement of these monetization constraints. DarkScan combines DOM analysis, text classification, visual detection, and dynamic behavioural flow tracing to identify manipulative patterns and quantify their behavioural and psychological impact (P et al., 2026, p. 2). Continuous integration pipelines for MindfulFeed should therefore include automated scans of all payment, upgrade, and consent screens, with severity scoring via models such as DPSS used to flag flows that unintentionally introduce obstruction or roach motel characteristics. Versioned flow diagrams for monetization pathways, as recommended for high risk sectors, can be stored alongside audit results to provide regulators and internal ethics bodies with traceable evidence of dark pattern free revenue

mechanisms (P et al., 2026, p. 8). Privacy and informed consent requirements in computational monitoring environments further constrain data based monetization. AI driven analytics over behavioural traces, communication logs, and psychological states risk misuse and discrimination without stringent safeguards, requiring transparent consent protocols that clearly explain what data are collected, how they are processed, and for what purposes, with consent iterative and revocable and data minimized to legitimate objectives (Sarkar & Kudapa, 2024, p. 18). If any paid features in MindfulFeed depend on enhanced analytics or personalised wellbeing recommendations, associated consent dialogues must adhere to this triad, avoiding complex privacy policies, confusing consent mechanisms, and defaults that encourage unintended disclosure, all of which are common in manipulative commercial designs (Sarkar & Kudapa, 2024, p. 18; T & R, 2026, p. 2). Algorithmic bias and fairness concerns in healthcare oriented AI monitoring show that monetizing advanced analytics without diverse data representation and fairness aware design can reproduce inequities across gender, race, age, or role, misclassifying stress levels or failing to detect vulnerability in underrepresented groups (Sarkar & Kudapa, 2024, p. 18). Consequently, MindfulFeed's monetization strategy cannot tie premium access to more accurate or fair recommendations for only paying users; fairness and accessibility must be treated as baseline obligations, with paid tiers restricted to non discriminatory enhancements such as additional reflection tools, provided that these are presented plainly and without scarcity cues or confirm shaming (Sarkar & Kudapa, 2024, p. 18; Mavi, 2025, p. 5127387). Finally, regulatory expectations around consent banners under GDPR and the ePrivacy Directive, including requirements that consent be freely given, specific, informed, unambiguous, and revocable, imply that revenue related flows cannot condition core access on intrusive tracking or opaque personalization (Gray et al., 2021, p. 4). Incorporating legal scholarship into design reviews means that every monetization related interface is assessed for preselected options, hidden opt outs, and asymmetric visual prominence, with redesign mandated wherever such features appear. In combination, dark pattern taxonomies, automated detection, privacy triads, and fairness obligations provide the evidence based constraints within which MindfulFeed's monetization remains compatible with pro social, cognitive health oriented objectives rather than undermining them for short term conversion gains (P et al., 2026, p. 2; Mavi, 2025, p. 2; Gray et al., 2021, p. 4; Sarkar & Kudapa, 2024, p. 18). The broader literature on the digital welfare state also cautions that large scale digitisation creates new vectors for exclusion, systemic error, and data driven control that may exacerbate inequality unless governance and oversight are explicitly designed to prevent such outcomes. These concerns about automated decision making and discrimination in public systems may suggest that platform monetization strategies should be treated with similar legal and ethical scrutiny (Yakoviyk & Tragniuk, 2024, p. 2).

10 Evaluation of MindfulFeed Across Desirability, Feasibility, and Profitability

10.1 User Desirability Assessment

User desirability for MindfulFeed depends on whether its interaction model, information architecture, and AI personalization features align with documented user expectations for autonomy, transparency, and cognitive relief rather than engagement maximization. Cognitive walkthroughs on major social media platforms show that dark patterns often appear in combinations across content engagement, account creation, and termination, generating distraction during learning and exposure to harmful content that undermines educational potential and lowers achievement. These harms suggest that desirability for a prosocial app is strongly conditioned on the absence of such combinations, particularly in navigation flows tied to attention and account control. A development record that explicitly maps and removes these pattern clusters, and documents low cognitive load in walkthroughs, directly addresses user demand for less manipulative engagement (Ukgoda, 2024, p. 3). Methodologically robust evidence on dark patterns demonstrates measurable impacts on decision quality, platform trust, and psychological wellbeing, with aggressive consent designs reducing privacypreserving choices by up to 23% and complex cookie banners architecturally determining withdrawal behaviour. For desirability assessment, this implies that users are likely to value MindfulFeed only if consent and configuration flows visibly diverge from such architectures. Testing protocols must therefore record whether participants can identify and enact privacypreserving choices without confusion or excessive effort, and whether they report higher trust when symmetric optin and optout pathways are implemented. Pos-

itive responses on these dimensions constitute direct evidence that design changes enhance perceived desirability relative to mainstream platforms (P et al., 2026, p. 2). Governance research on digital wellbeing introduces four analytical dimensions that link administrative design with experiential outcomes: accountability, transparency, oversight, and wellbeing outcomes. Desirability assessment for MindfulFeed can be structured around these dimensions. Interview and questionnaire instruments should capture whether users can locate accountability cues (for example, explanations of who is responsible for algorithmic decisions), understand how AI systems affecting cognition and emotion operate, identify mechanisms for feedback and correction, and describe emotional stability, cognitive autonomy, and social connection when using the app. Credibility, transferability, and dependability in such qualitative evaluation depend on triangulating user narratives, expert walkthrough findings, and interaction logs, mirroring NVivosupported thematic content analysis procedures used in governance studies (Sharma, 2025, p. 7808872). High scores across these four dimensions, combined with robust qualitative coding, indicate that the product's value proposition aligns with expressed user needs for ethically governed digital experiences. Mindfulness app research employing heuristic evaluation, survey, diary, and interviews over nine weeks reveals that commercialization and profit motives can lead to manipulative design techniques that contradict the foundations of mindfulness. Brignull's categories such as bait and switch, confirm shaming, disguised ads, forced continuity, friend spam, and hidden cost illustrate how apps can pressure users into unwanted subscriptions, guiltdriven confirmations, or hidden charges. Findings from this multimethod study show that such dark patterns negatively shape user perceptions and engagement, including discontinuation and "fleeing" behaviour. For MindfulFeed, desirability assessment must therefore include heuristic reviews of monetization and habitforming features, diaries capturing reasons for nonusage, and interviews probing reactions to any persuasive element. Reduced incidence of guilt, confusion, or perceived exploitation compared with the documented mindfulness app patterns would provide strong evidence that MindfulFeed's ethical design increases desirability among users seeking mental health support tools (Özkürkçü & Doğanay, 2025, p. 2). Systematic review work on AIbased burnout interventions demonstrates the value of structured coding frameworks, MMAT quality assessment, and narrative synthesis for understanding how computational methods affect stress and burnout outcomes. Translating these procedures into desirability evaluation, MindfulFeed's testing data should be coded for variables such as perceived cognitive load, stress change during use, clarity of AI explanations, and perceived usefulness of personalization controls. Interrater reliability measures like Cohen's kappa can be used to ensure consistent coding of desirability indicators across evaluators (Sarkar & Kudapa, 2024, p. 19). Evidence indicates that users appreciate AI systems that enhance information acquisition efficiency and satisfy personal needs when recommendation accuracy, transparency, and reliability are high, and when trust and perceived usefulness are correspondingly strong (Zhu et al., 2026, p. 6). Questionnaire measures adapted from these constructs can quantify whether MindfulFeed's recommendation features increase behavioural intention to accept and continuously use content suggested in a healthsupport context, without undermining autonomy. Taken together, desirability assessment for MindfulFeed must combine taskbased expert walkthroughs, multimethod user studies, and structured coding of trust, autonomy, and cognitive strain. Only if empirical indicators show reduced dark pattern exposure, enhanced governance visibility, and sustained perceived usefulness of transparent AI recommendations can the product be considered desirable in light of current evidence on social media harms and ethical digital wellbeing (P et al., 2026, p. 2; Ukgoda, 2024, p. 3; Sharma, 2025, p. 7808872; Zhu et al., 2026, p. 6).

10.2 Technical and Organizational Feasibility

Technical and organizational feasibility for MindfulFeed hinges on whether the app's governance, data, and AI practices can be implemented within existing institutional capabilities and regulatory expectations. Evidence from occupational health monitoring shows that computational systems routinely handle highly sensitive behavioural and psychological data, yet responsible deployment is achievable when privacy governance follows a triad of informed consent, data minimization, and transparent communication between institutions, employees, and system designers. Translating this to a social app context implies that feasible implementation requires technical infrastructure to manage iterative and revocable consent states, strict data minimization at collection and storage layers, and auditable communication channels explaining how recommendation and analytics components operate. Embedding algorithmic transparency and auditability into MindfulFeed's architecture is also feasible given current AI practices. In healthcare monitoring, ethical integrity demands that predictive systems

be explainable and auditable, allowing users to understand how conclusions about mental health are reached and preventing AI from being perceived as an intrusive management tool (Sarkar & Kudapa, 2024, p. 18). Practically, this necessitates logging of model inputs and outputs, traceable configuration of recommendation parameters, and interfaces capable of surfacing "why this content" explanations. Similar demands appear in explainable AI and governance work in quantitative finance, where systems are expected to trade off some opacity for pipelines that are explainable and enhanced with domain knowledge so that stakeholders can understand, audit, and maintain confidence in AI strategies (Guo, 2025, p. 4). MindfulFeed's technical feasibility therefore depends on adopting explainability modules, governance hooks, and data provenance verification routines that are already recognised as achievable in other AI domains. Organizational feasibility is shaped by the need for continuous ethical governance rather than oneoff compliance. In AIenabled occupational health, ethical governance is described as a precondition for responsible and humane application of AI, requiring organizational cultures that prioritise compassion, transparency, and trust over control and surveillance, alongside internal structures such as AI ethics boards and compliance committees to oversee data handling, model validation, and risk management (Sarkar & Kudapa, 2024, p. 18). For MindfulFeed, this implies that feasible deployment depends on instituting comparable oversight bodies, integrating ethical review of dark patterns and consent flows into design governance, and aligning internal policies with privacy, fairness, and accountability requirements. Systematic review methodology in computational burnout research demonstrates that organizations can manage complex, technologyheavy projects with structured processes. The review employed PRISMA guidelines, multidatabase searches, staged identification and screening, structured coding frameworks, thematic clustering, and quality assessment via the Mixed Methods Appraisal Tool, with interrater reliability measured by Cohen's kappa (Sarkar & Kudapa, 2024, p. 19). These procedures show that institutions can sustain methodologically rigorous, dataintensive workflows when roles, tools, and quality thresholds are clearly defined. MindfulFeed's development process can feasibly mirror these structures in its Empathize, Define, Ideate, Prototype, and Test phases, for example by treating usability logs and darkpattern audits as "studies" that pass through scripted identification, coding, and inclusion stages. Responsible digital education analysis adds further organisational guidance. Institutions are advised to apply pedagogy first, equity by design, privacy by design, and evidence before scale, with data minimization, human oversight, explainability, impact assessment, correction rights, and security implemented before using AI in important decisions (Judijanto, 2026, p. 15). This indicates that MindfulFeed is organizationally feasible only if its governance model adopts similar principles: specifying wellbeing and autonomy objectives before technology choices, involving vulnerable groups in design, limiting data from the start, and scaling features only after careful evaluation of results, risks, and costs. Finally, darkpattern research provides concrete technical standards and audit requirements that MindfulFeed can realistically adopt. Designers are advised to implement symmetric design principles so that opting out, cancelling, or exercising privacy preferences are no more complex than their inverses, to use plain language in calls to action, and to conduct regular thirdparty audits using automated tools such as DarkScan within continuous integration pipelines. Organizations are further encouraged to maintain user journey documentation for key pathways and treat user trust, task success, and longterm retention as success metrics alongside conversion, while regulators move toward mandatory "darkpatternfree" certification in highrisk sectors (P et al., 2026, p. 8). These recommendations show that technical artifacts such as automated scans, flow documentation, and trust metrics are practically implementable and form a feasible backbone for MindfulFeed's organisational commitment to darkpattern avoidance and cognitivehealth protection.

10.3 Profitability and Business Sustainability Analysis

Profitability and longterm business sustainability for MindfulFeed must be evaluated within constraints set by empirical work on dark patterns, privacy, and ethical AI governance. Research on manipulative interfaces shows that many revenueoptimizing designs rely on strategies such as sneaking, urgency or scarcity cues, obstruction, misdirection, forced action, confirm shaming, roach motel flows, bait and switch, and drip pricing, all purposefully crafted to steer users into actions they did not plan to take by exploiting default bias, scarcity bias, framing, and loss aversion (Mavi, 2025, p. 2). Experimental analysis demonstrates that complex cancellation pathways, consisting of multiple forced detours and confirmation hurdles, generate high abandonment rates and extended task times, while aggressive dark patterns in consent and checkout contexts can dramatically increase subscription uptake compared with neutral designs. For MindfulFeed, these findings imply that many conventional optimization levers for

profitability are ethically unavailable; financial viability must instead be grounded in value propositions that do not depend on deceptive friction or emotionally manipulative copy. Organizational recommendations derived from dark pattern severity assessment and automated detection pipelines such as DarkScan provide a blueprint for sustainable, nonexploitative business practices. Designers are advised to adopt symmetric design principles so that opting out, cancelling, or exercising privacy preferences is no more complex than the corresponding optin flows, and platform operators are encouraged to align product success metrics with user trust scores, task success rates, and longterm retention rather than shortterm conversion alone. These prescriptions can be operationalized in MindfulFeed's business dashboards by prioritising metrics such as sustained daily active use without escalation of engagement, completion of consent and configuration tasks without confusion, and stable retention over time, accepting lower immediate conversion to paid features as the tradeoff for reputational resilience and compliance readiness. Publishing transparency reports on optimization techniques and maintaining full user journey documentation for key flows are further recommended as organisational practices that support regulatory scrutiny and trust (P et al., 2026, p. 8). Embedding these artefacts into the product development record contributes directly to business sustainability by reducing future compliance risks and fostering user confidence. Privacy and AI governance research underscores that business models built on extensive behavioural surveillance and opaque personalization face structural risk. Contemporary platforms often employ complex privacy policies and confusing consent mechanisms that encourage unintended data disclosure, limiting users' understanding of how data are collected, processed, and shared (T & R, 2026, p. 3). As personalization technologies become more sophisticated, they may amplify manipulative practices by tailoring deceptive interfaces to individual characteristics, prompting calls for stronger ethical guidelines and regulatory frameworks that prioritize welfare alongside business objectives (T & R, 2026, p. 3; Ronald, 2025, p. 7). MindfulFeed's profitability strategy must therefore avoid monetization structures that depend on largescale, finegrained profiling without robust consent and transparency, because such structures are increasingly subject to regulatory challenge and reputational backlash. Instead, sustainable revenue can be tied to clearly delimited, consented analytics that support preventive wellbeing interventions, where machine learning models analyze patterns to identify concerns before they become severe, provided that monitoring remains proportionate and rightsrespecting (Ronald, 2025, p. 7). Systematic review methodology in computational burnout interventions suggests that organisations can manage complex, AIenabled services with high analytical rigor when structured processes for monitoring and evaluation are in place. PRISMA-guided reviews rely on staged identification, screening, eligibility, and inclusion, structured coding frameworks, quality appraisal tools, and interrater reliability checks to maintain integrity. Translating this into ongoing business operations, MindfulFeed can support sustainability by instituting analogous internal review cycles for new features and monetization changes, screening proposed revenue mechanisms against inclusion criteria that cover ethical transparency, data minimization, and user wellbeing impacts. Features failing quality thresholds would be excluded before launch, lowering the probability of costly post hoc remediation (Sarkar & Kudapa, 2024, p. 19). Continuous ethical oversight of recommendation and consent components, aligned with governance dimensions of accountability, transparency, oversight, and wellbeing outcomes, further reduces longterm legal and social risk, which is a central component of business sustainability in digital health and wellbeing sectors (Sharma, 2025, p. 8; Ronald, 2025, p. 7). Finally, studies on consumer awareness show that exposure to dark patterns is high but comprehension remains low, with many users misclassifying manipulative designs as mere bad design and knowing little about datadriven consent manipulation (Mavi, 2025, p. 2). While such low awareness may temporarily permit exploitative monetization elsewhere, aligning MindfulFeed with emerging regulatory standards that restrict deceptive consent practices and move toward mandatory darkpatternfree certification in highrisk domains positions the app advantageously in the long term (Gray et al., 2021, p. 4; P et al., 2026, p. 8). Profitability, in this evidencebased frame, becomes linked to user trust, regulatory compatibility, and demonstrable protection of autonomy, which collectively underpin durable business sustainability for an ethical, prosocial social platform.

11 Conclusion

This work synthesises empirical research on dark patterns, behavioral economics, and governance to articulate a coherent design and development agenda for MindfulFeed. It frames digital wellbeing through four operational dimensions, accountability, transparency, oversight, and wellbeing outcomes,

and translates those into concrete constraints for interface design, algorithmic personalization, consent flows, and monetization. Prototypes progressing from a lowfidelity batch feed to a midfidelity algorithm control dashboard and a highfidelity interactive system demonstrate how symmetric effort in optin/optout, plain language explanations, tiered personalization, fatigueaware UI nudges, and auditable model records can reduce manipulation, preserve cognitive autonomy, and maintain compliance with regulatory regimes such as GDPR and ePrivacy. Testing protocols adapted from cognitive walkthroughs, systematic review practices, and automated detection tools provide a repeatable evaluation pipeline that flags residual pattern risks, measures decision quality and trust, and records governance artefacts for future audit.

Operational recommendations emphasise embedding human oversight, iterative and revocable consent, data minimization, fairness audits, and accessible explanation mechanisms into both product and organizational processes. Monetization strategies are constrained to avoid tactics that exploit bias or friction, instead relying on transparent value exchange and non discriminatory enhancements while accepting different shortterm conversion dynamics in exchange for reputational and regulatory resilience. By documenting design rationales, mapping features to governance pillars, and integrating automated darkpattern scanning into development workflows, this approach offers a practicable pathway for building social platforms oriented to sustainable digital wellbeing and measurable protection of attention, emotion, and autonomy.

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